

Final Project

Predict the average reaction time through the variables we get from the survey

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## Warning: package 'leaps' was built under R version 4.3.3
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## Warning: package 'olsrr' was built under R version 4.3.3
```

```
## Warning: package 'glmnet' was built under R version 4.3.3
```

```
## Warning: package 'lmtest' was built under R version 4.3.3
```

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## Warning: package 'zoo' was built under R version 4.3.3
```

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## Warning: package 'nlme' was built under R version 4.3.3
```

```
## Warning: package 'car' was built under R version 4.3.3
```

```
## Warning: package 'carData' was built under R version 4.3.3
```

```
## [1] "First"          "Second"          "Third"           "Class"
## [5] "Age"            "AvgSleepTime"    "LastNightSleep"  "HoursAweak"
## [9] "Fatigue"        "Stress"          "Distraction"     "NoiseLevel"
## [13] "Temperature"    "VideoGamePlay"   "SportsPlay"      "ShootingGames"
## [17] "AvgHoursExercise" "Gamer"           "CaffeinIntake"   "AlcoholIntake"
## [21] "VisualAcuity"   "PrimarilyHand"   "UsePrimaryHand"  "Cautious"
## [25] "InputDevice"    "RefreshRate"     "DeviceOS"        "WiFi"
```

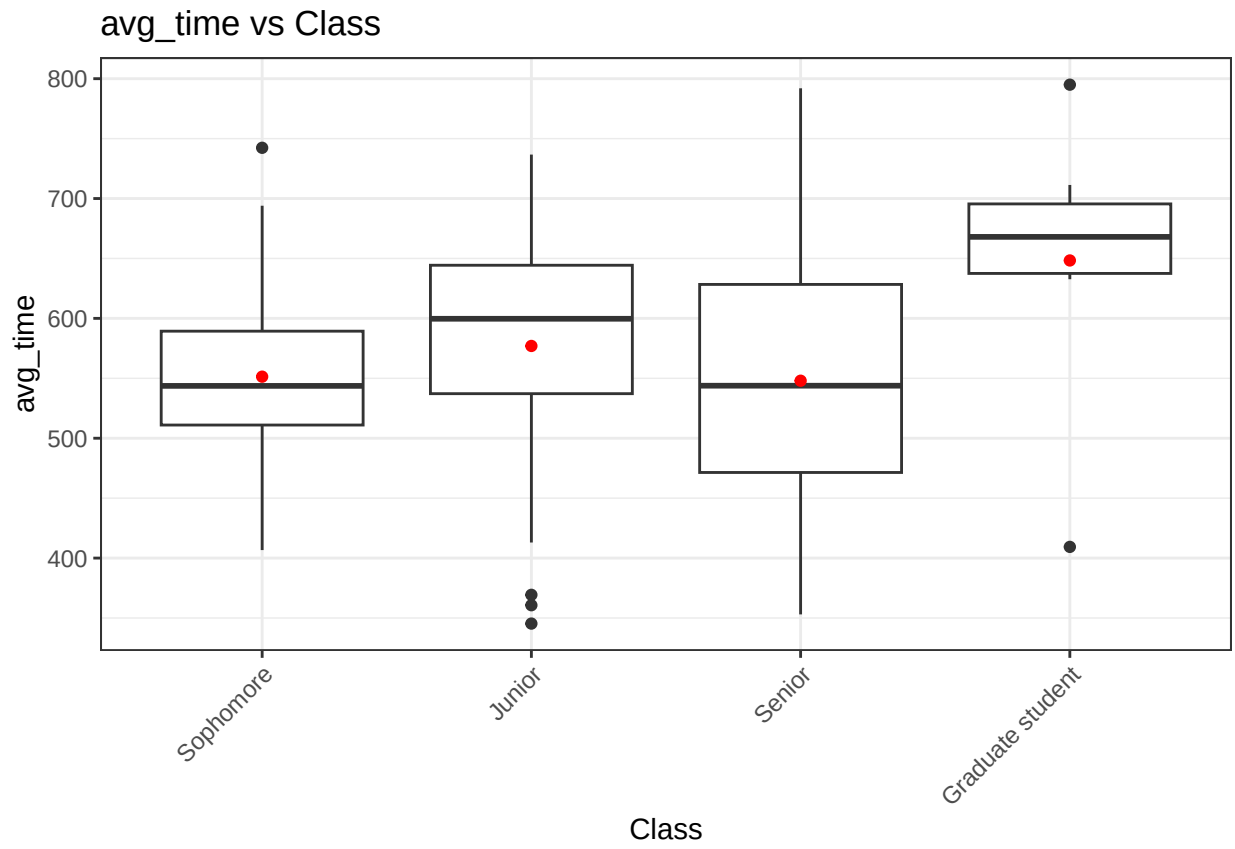
```
##      First      Second      Third      Class
## Min.   :325.0  Min.   :338.0  Min.   :341.0  Sophomore    :45
## 1st Qu.:520.0  1st Qu.:498.0  1st Qu.:490.0  Junior       :55
## Median :593.0  Median :561.0  Median :542.0  Senior       :22
## Mean   :592.5  Mean   :560.7  Mean   :547.7  Graduate student: 7
## 3rd Qu.:663.0  3rd Qu.:633.0  3rd Qu.:613.0
## Max.   :934.0  Max.   :833.0  Max.   :783.0
##      Age      AvgSleepTime  LastNightSleep  HoursAweak
## Min.   :18.0  Min.   : 5.000  Min.   : 4.000  Min.   : 0.000
## 1st Qu.:20.0  1st Qu.: 6.500  1st Qu.: 6.000  1st Qu.: 4.000
## Median :20.0  Median : 7.000  Median : 7.000  Median : 6.000
## Mean   :20.9  Mean   : 7.236  Mean   : 7.264  Mean   : 7.322
## 3rd Qu.:21.0  3rd Qu.: 8.000  3rd Qu.: 8.000  3rd Qu.:10.000
## Max.   :26.0  Max.   :10.000  Max.   :11.000  Max.   :18.000
##      Fatigue      Stress      Distraction      NoiseLevel
```

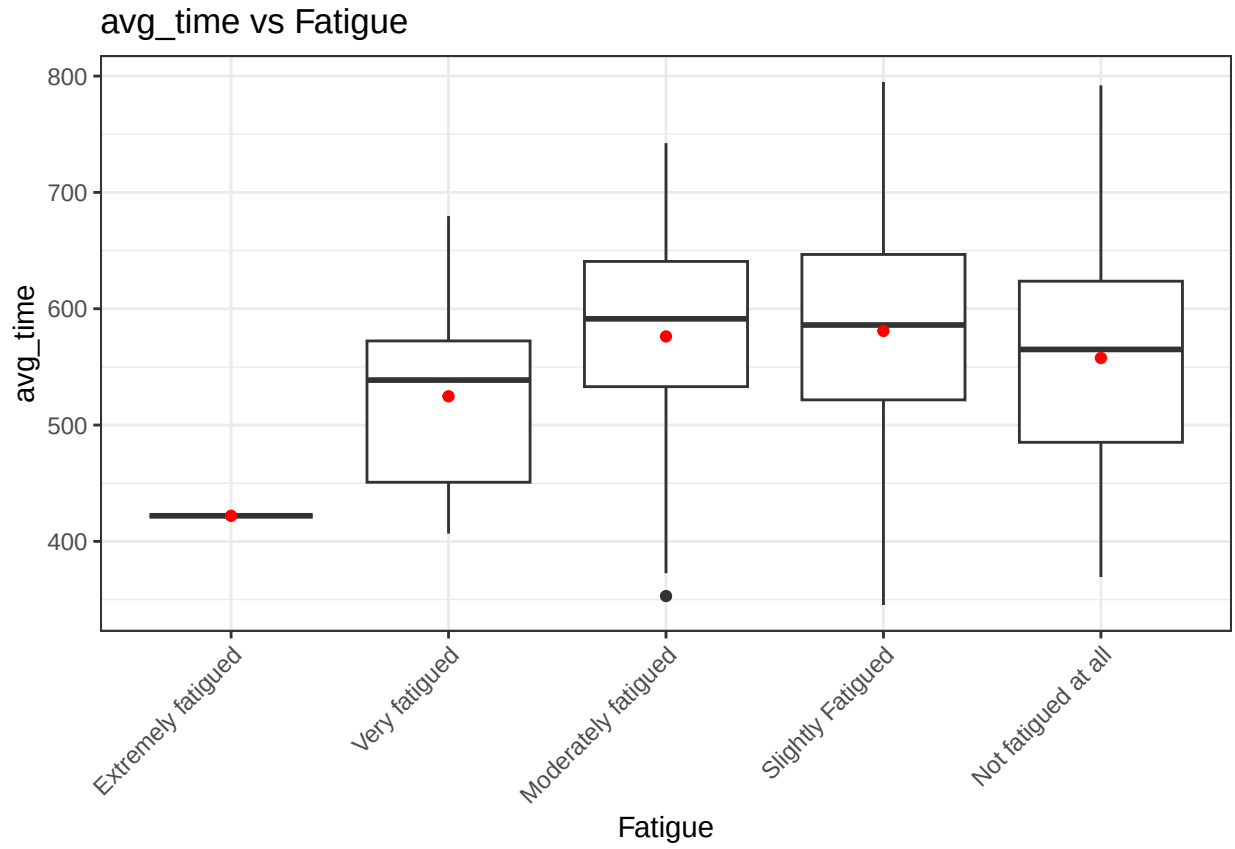
```

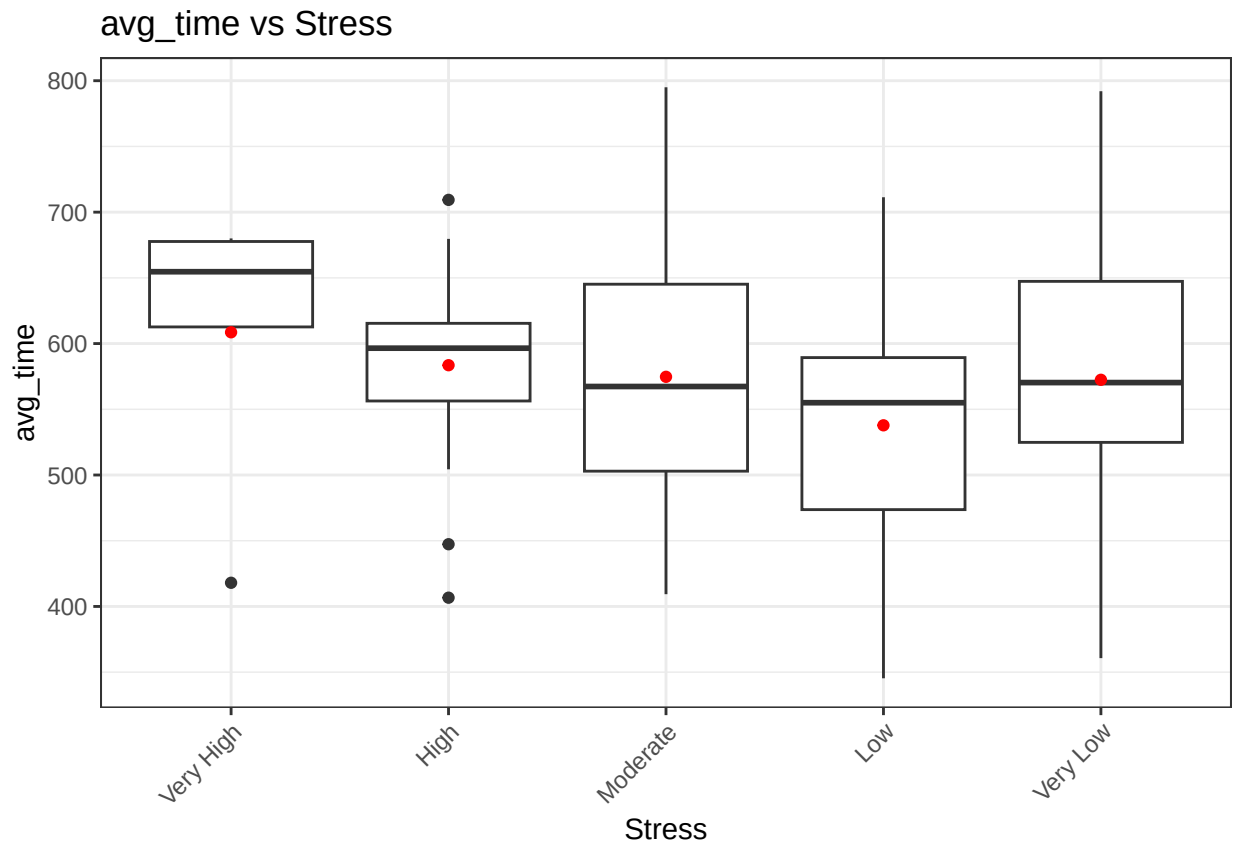
## Extremely fatigued : 1 Very High: 5 Length:129 Min. : 1.000
## Very fatigued :16 High :20 Class :character 1st Qu.: 2.000
## Moderately fatigued:44 Moderate :46 Mode :character Median : 3.000
## Slightly Fatigued :45 Low :35 Mean : 3.287
## Not fatigued at all:23 Very Low :23 3rd Qu.: 5.000
## Max. :10.000
## Temperature VideoGamePlay SportsPlay
## Very Warm: 2 Daily :11 Daily : 5
## Warm :16 Several times a week :36 Several times a week :24
## Neutral :98 Once a week : 9 Once a week :11
## Cold :13 Several times a month:24 Several times a month:24
## Rarely :38 Rarely :50
## Never :11 Never :15
## ShootingGames AvgHoursExercise Gamer CaffeinIntake
## None :19 Min. : 0.000 Length:129 Length:129
## Beginner :54 1st Qu.: 2.000 Class :character Class :character
## Intermediate:35 Median : 4.000 Mode :character Mode :character
## Advanced :18 Mean : 4.097
## Expert : 3 3rd Qu.: 6.000
## Max. :12.000
## VisualAcuity PrimarilyHand UsePrimaryHand
## Length:129 Right hand :119 Yes:122
## Class :character Left hand : 6 No : 7
## Mode :character Ambidextrous (both hands equally): 4
##
##
## Cautious InputDevice RefreshRate
## Extremely cautious : 3 Game controller: 1 High :29
## Very cautious :32 Mouse :47 Mid-Range:52
## Moderately cautious:68 Keyboard : 2 Standard :48
## Slightly cautious :21 Trackpad :69
## Not cautious at all: 5 Touch screen :10
##
## DeviceOS WiFi avg_time
## Length:129 Unstable: 2 Min. :345.3
## Class :character Stable :127 1st Qu.:511.0
## Mode :character Median :574.0
## Mean :567.0
## 3rd Qu.:635.0
## Max. :795.0

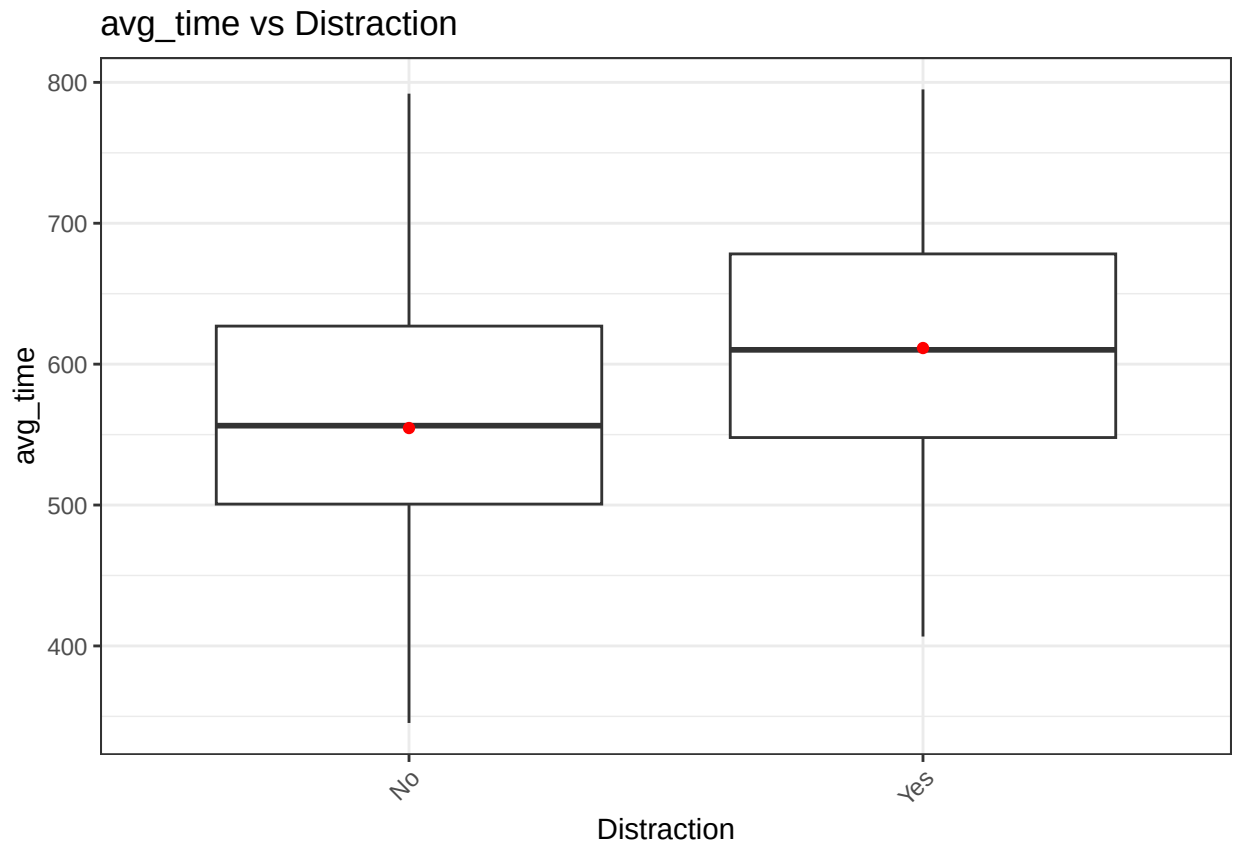
## Warning: 'aes_string()' was deprecated in ggplot2 3.0.0.
## i Please use tidy evaluation idioms with 'aes()'.
## i See also 'vignette("ggplot2-in-packages")' for more information.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

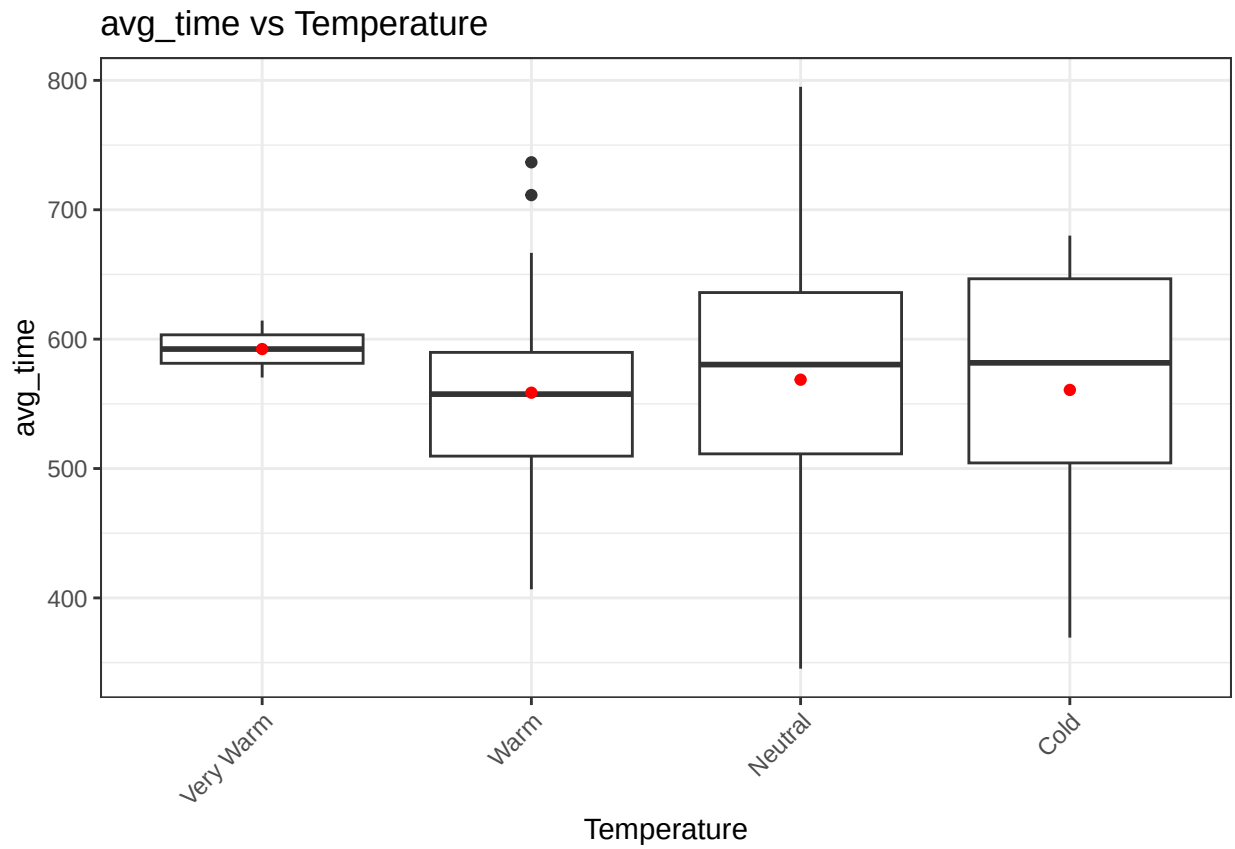
```

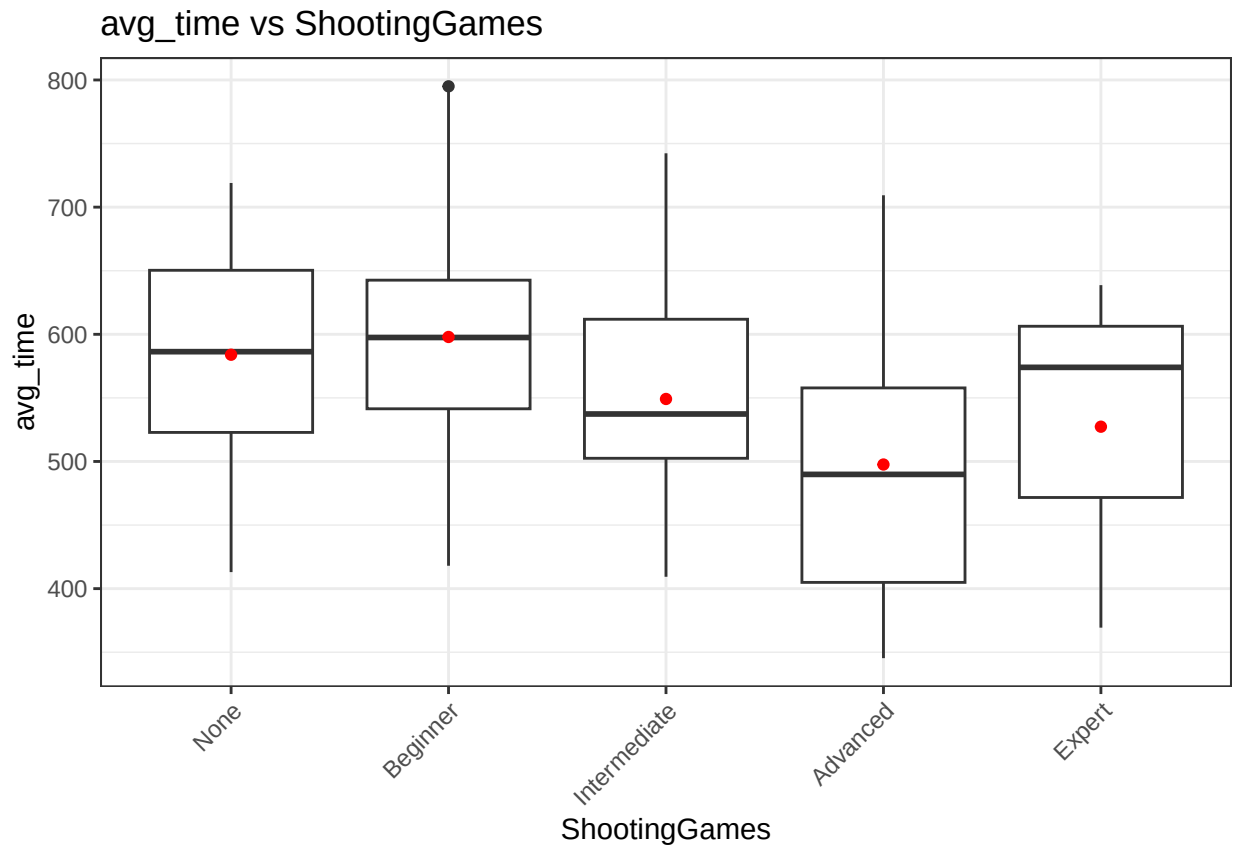


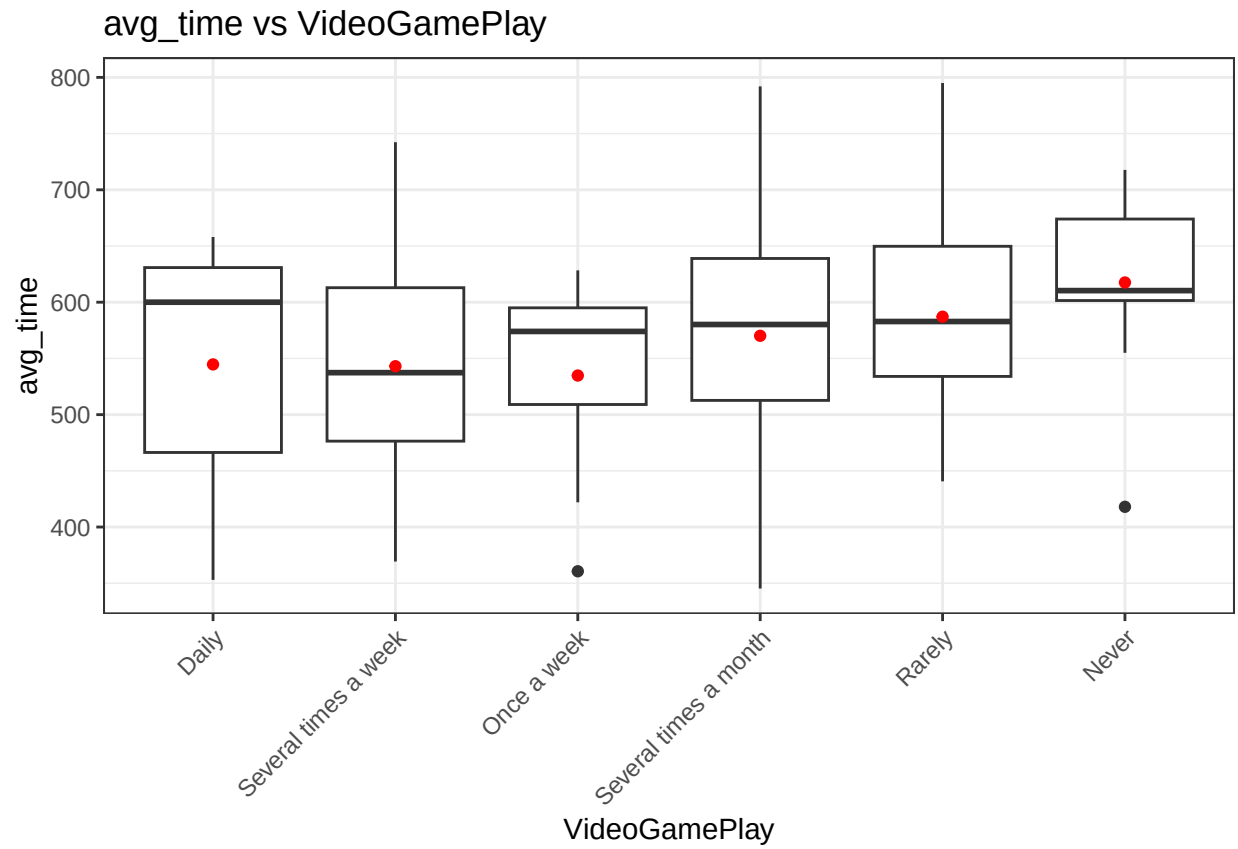


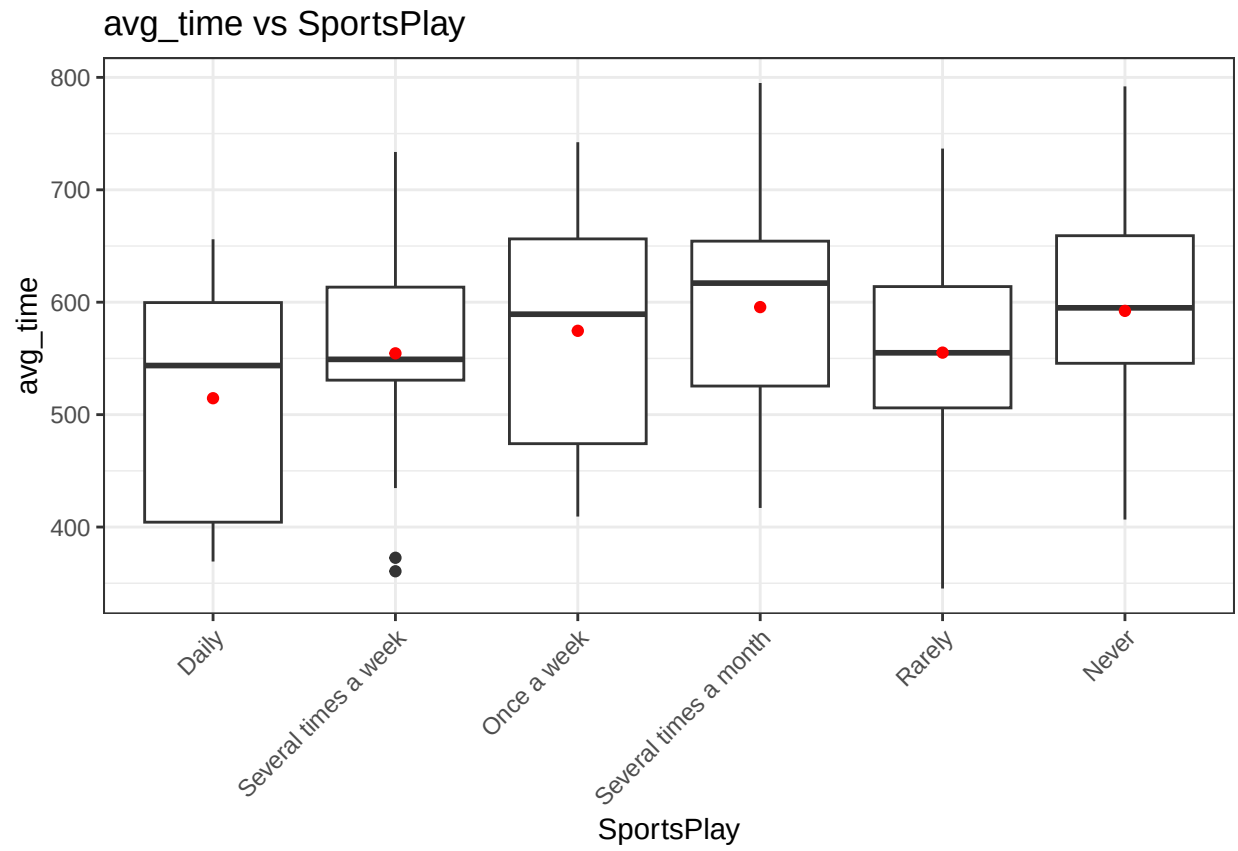


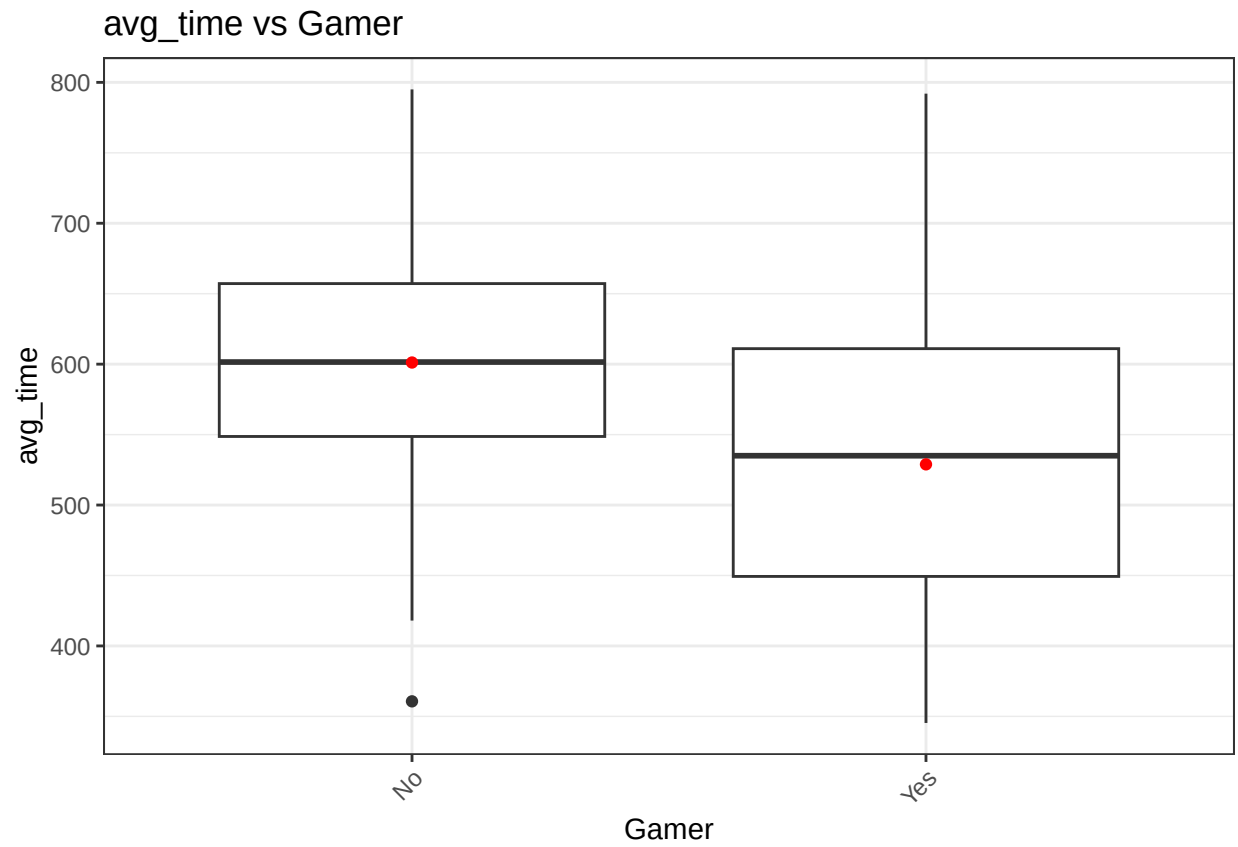


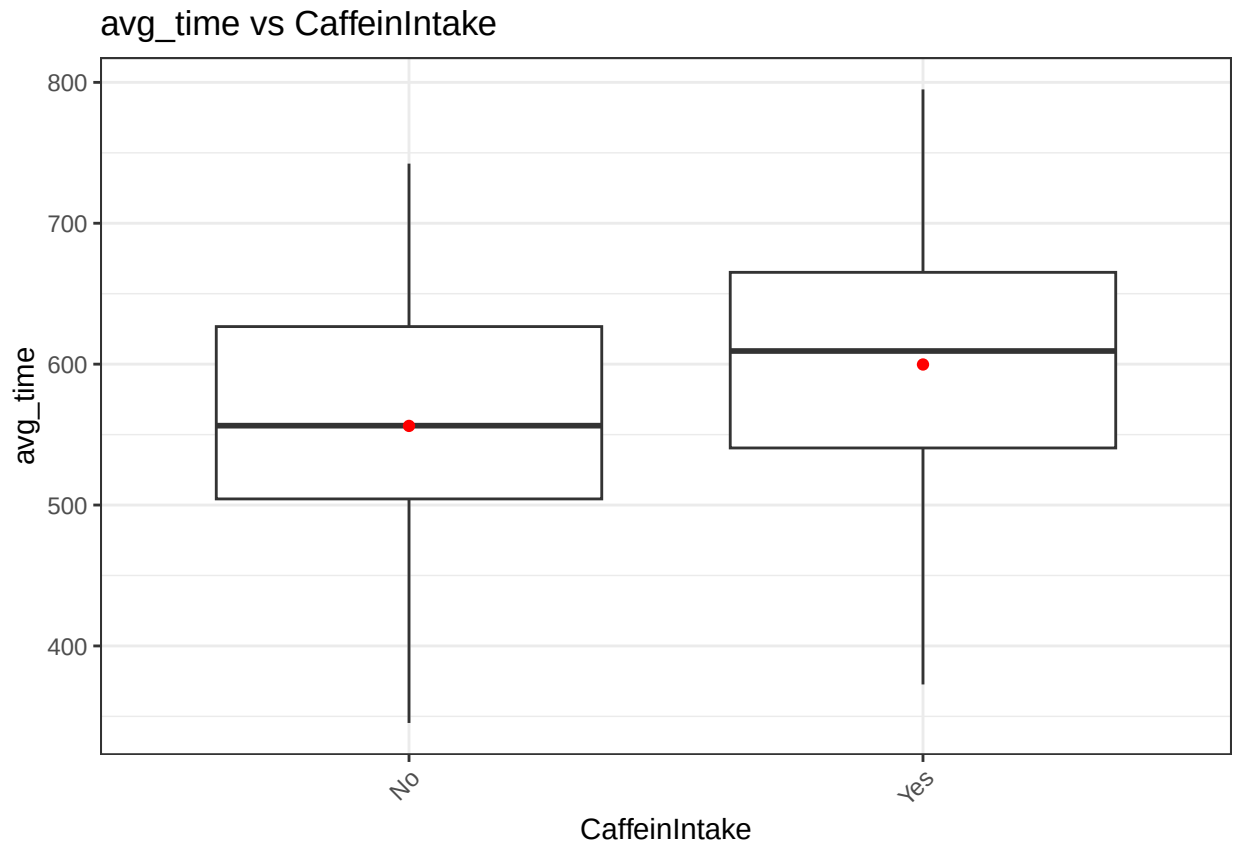


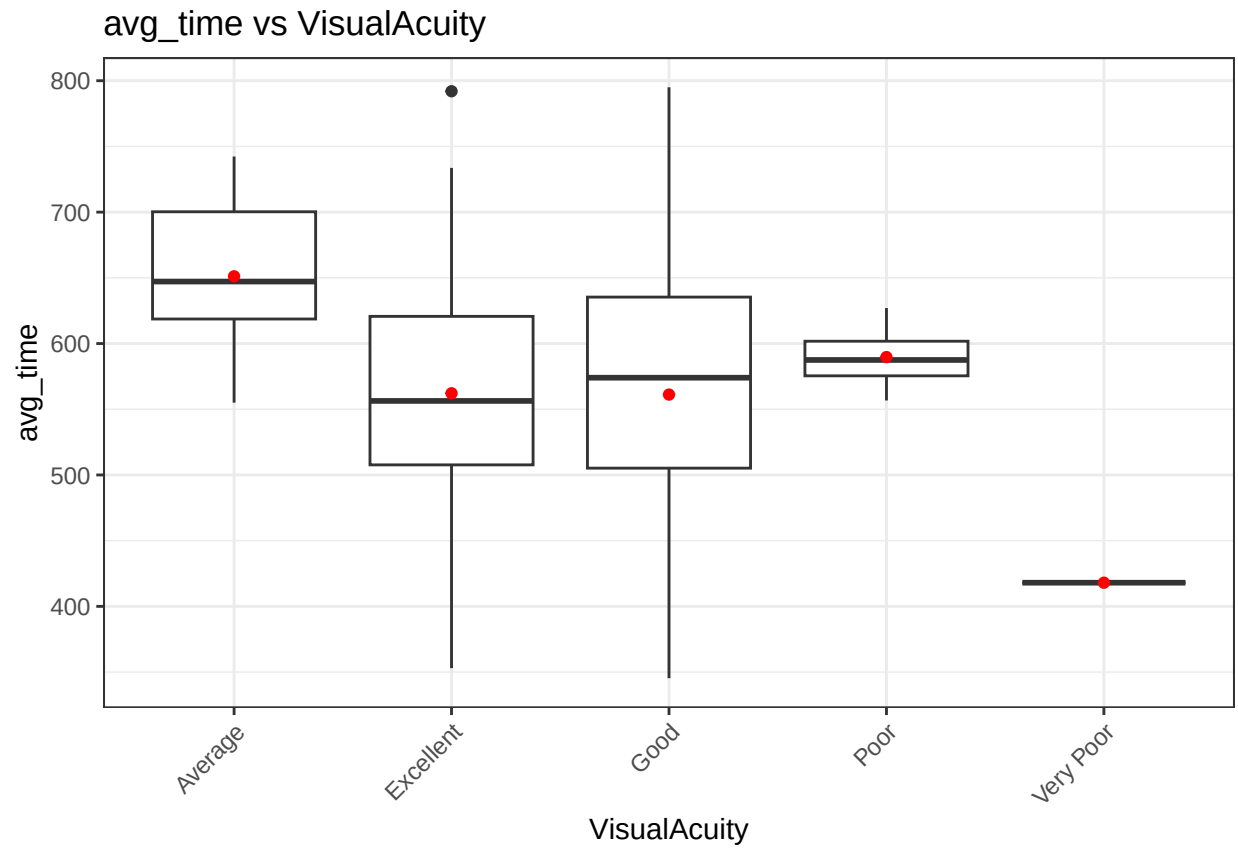


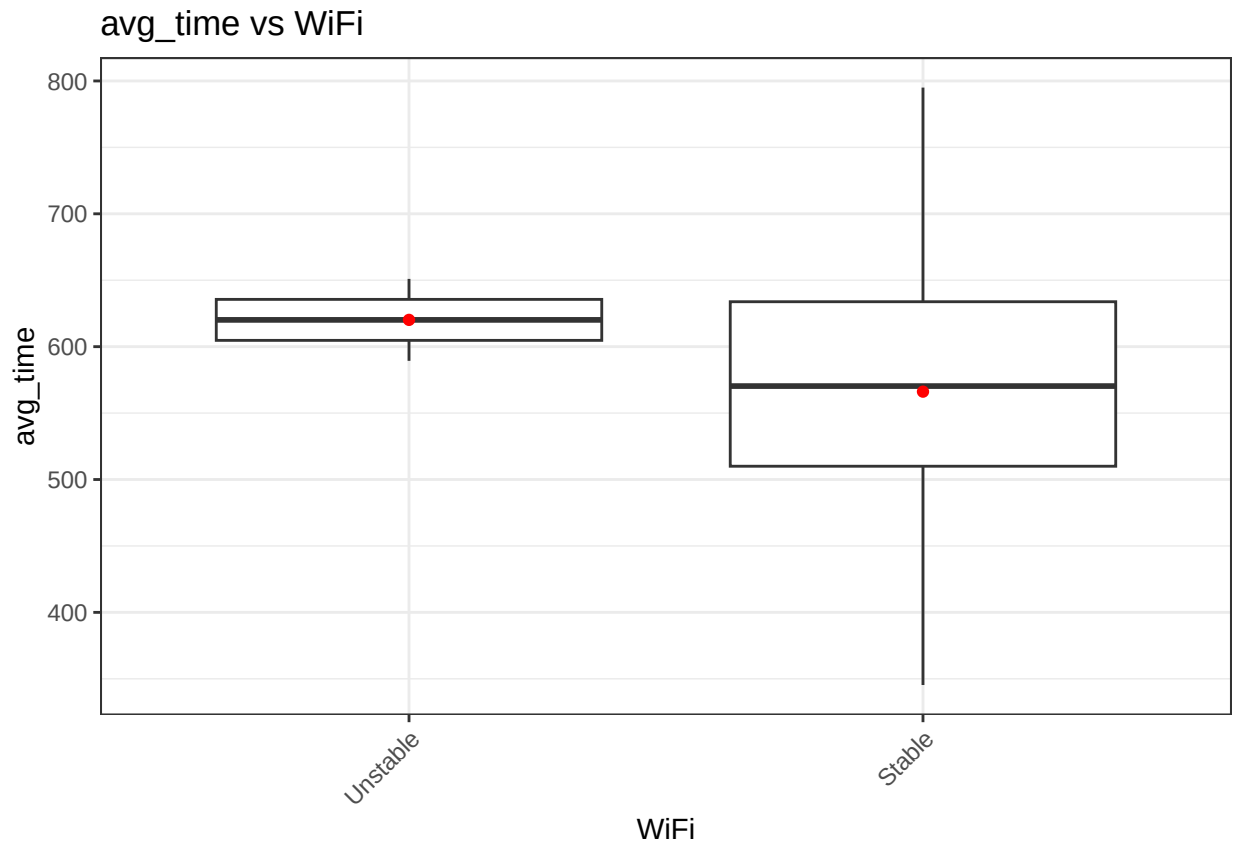


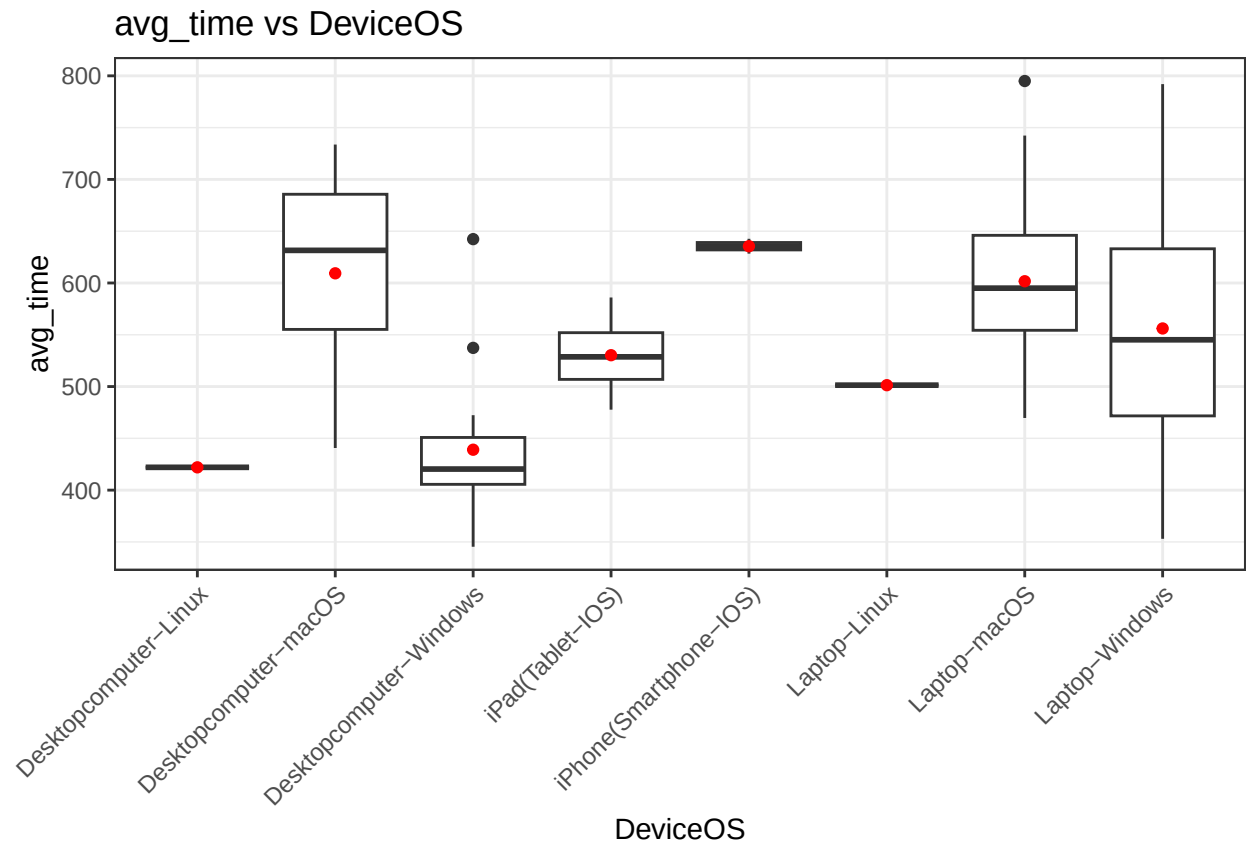


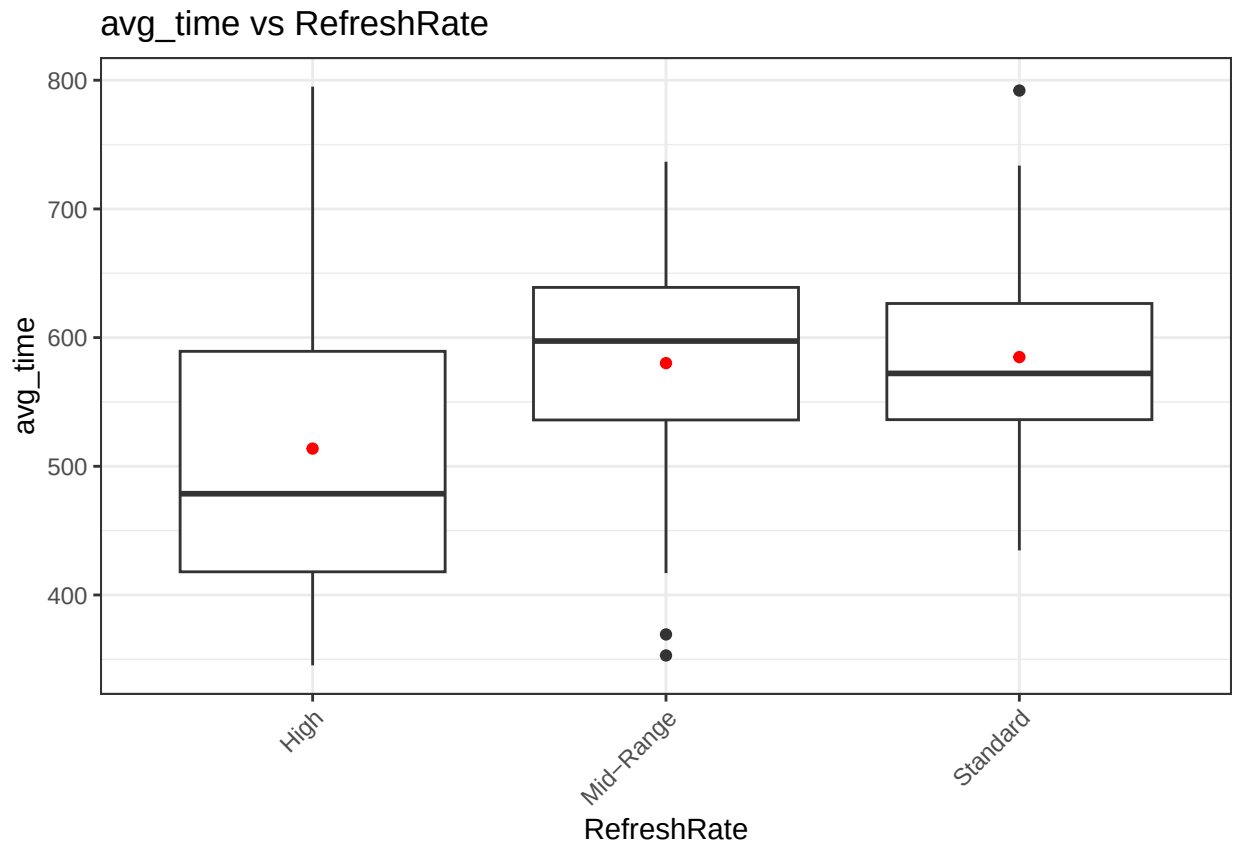


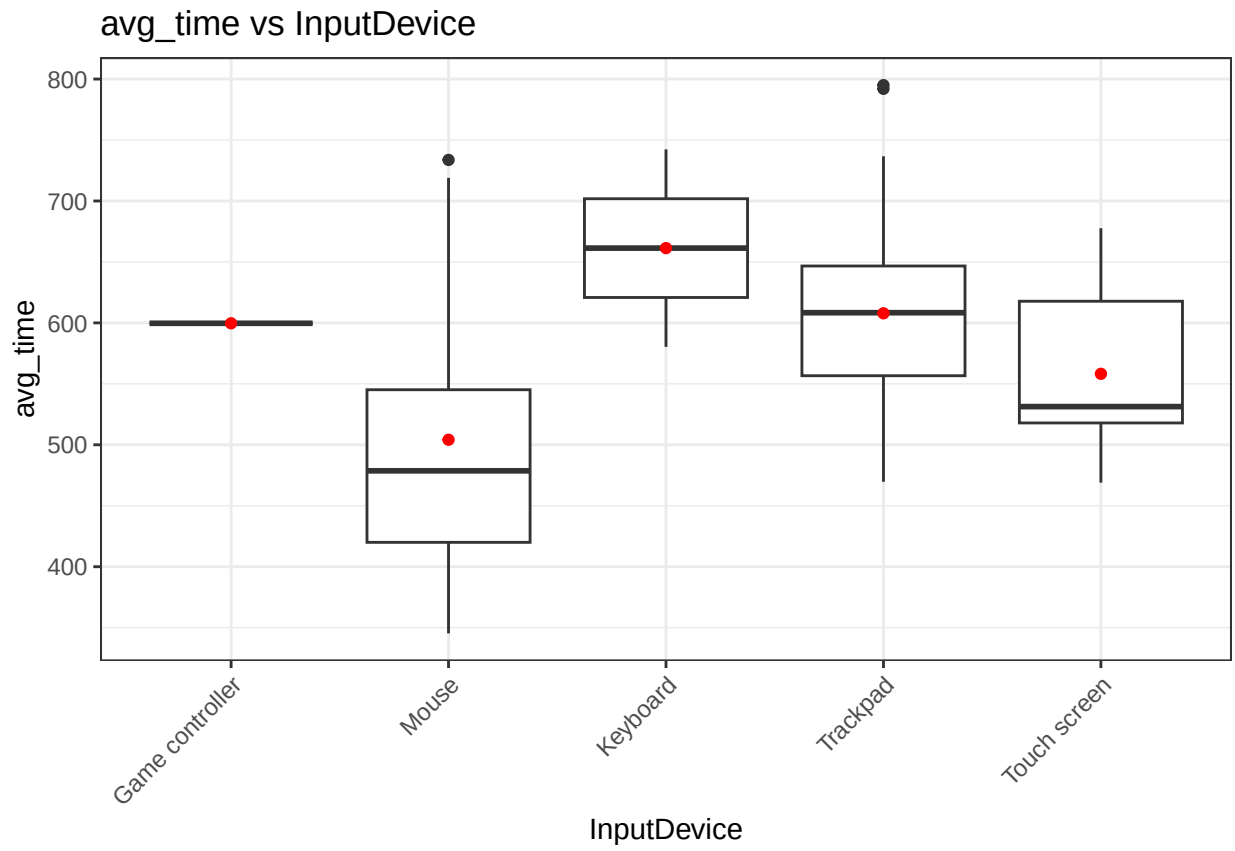


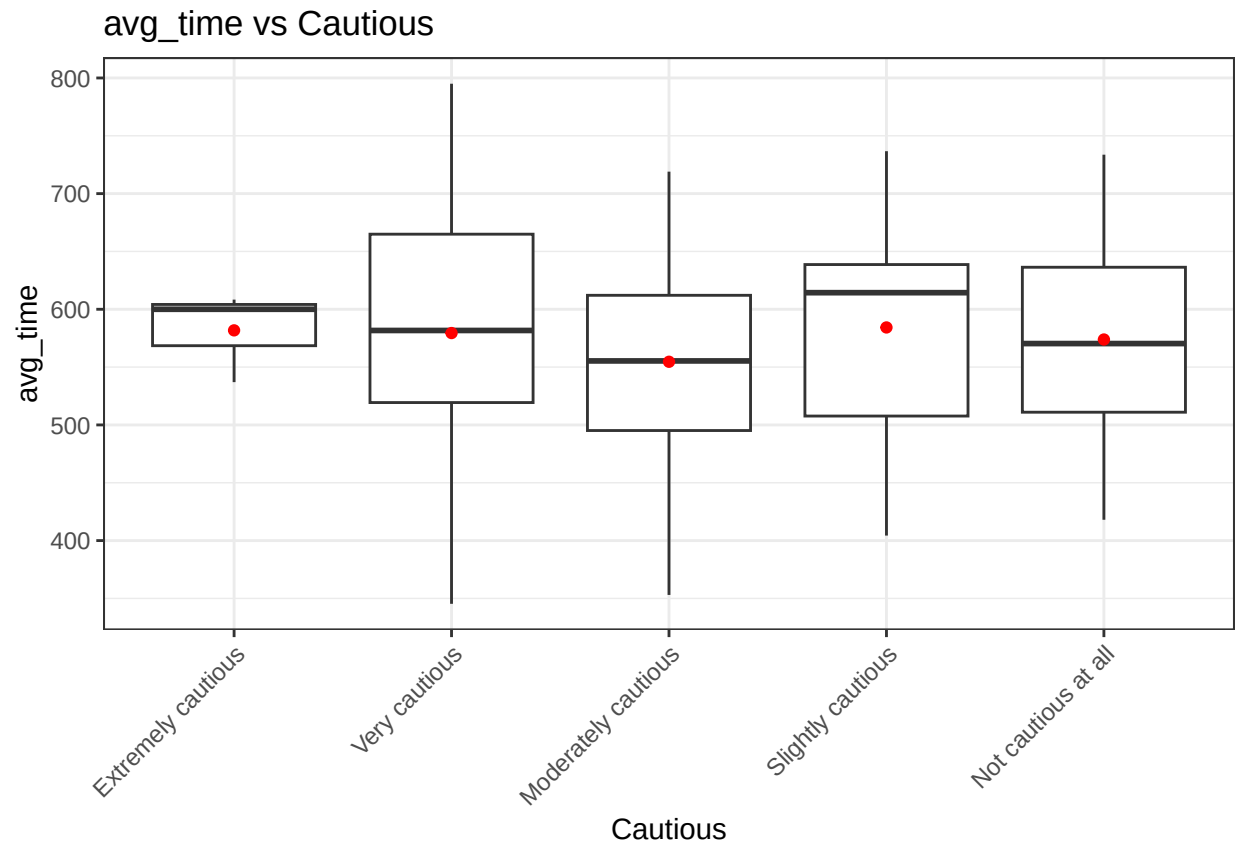


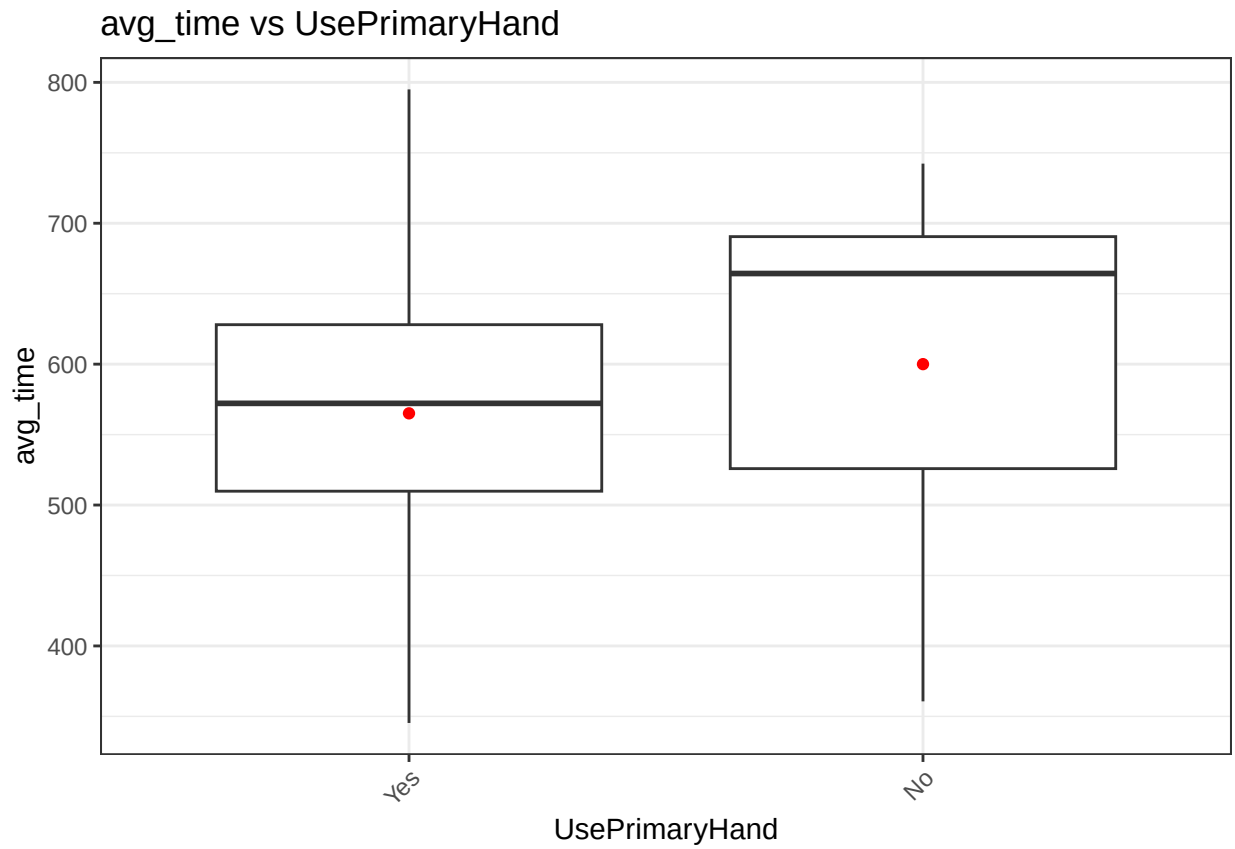


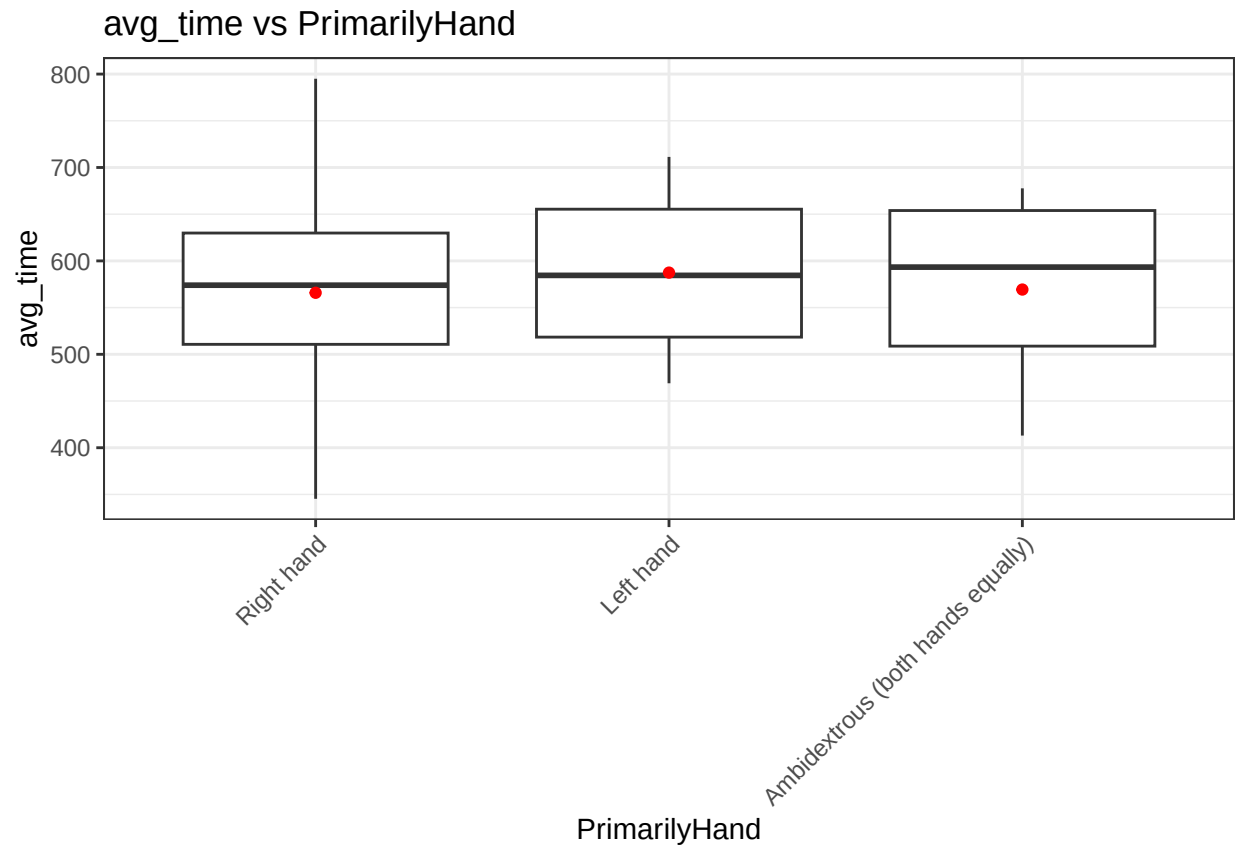


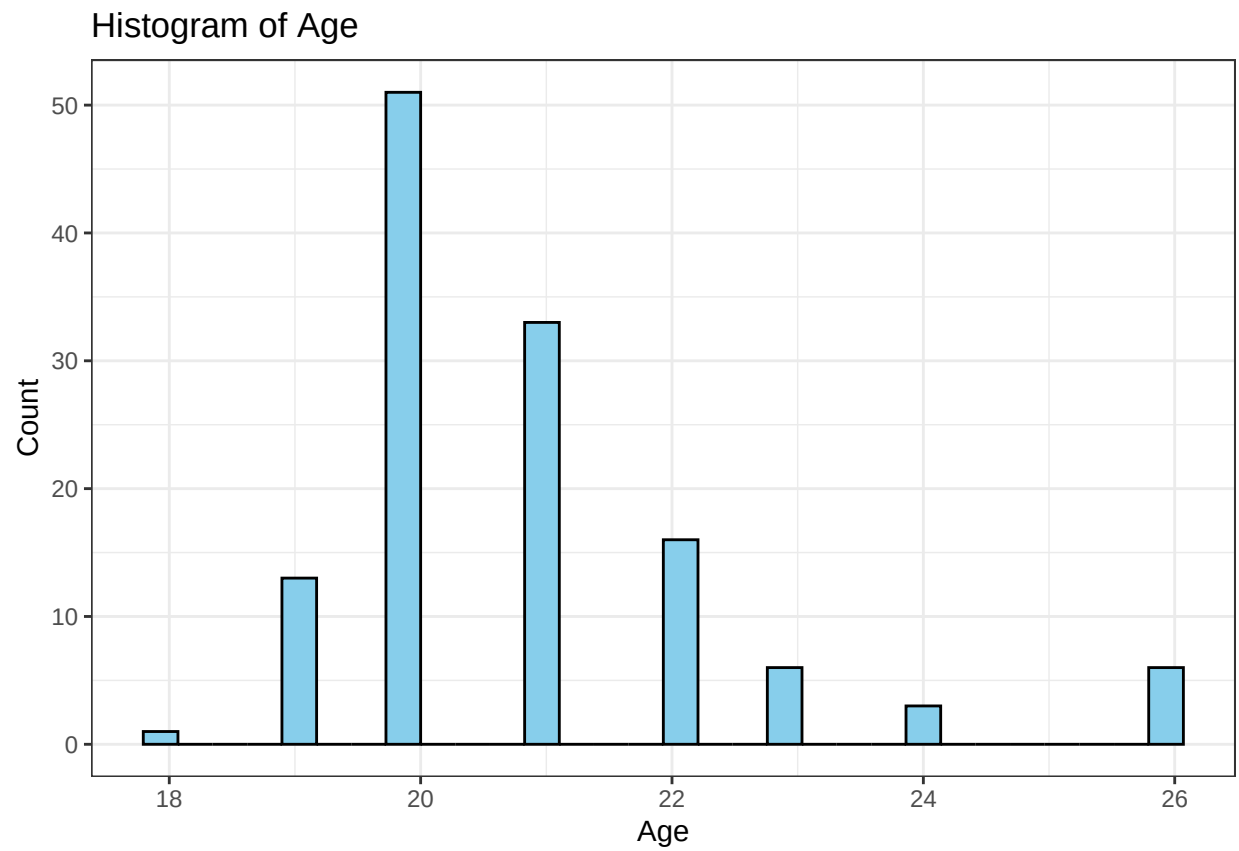


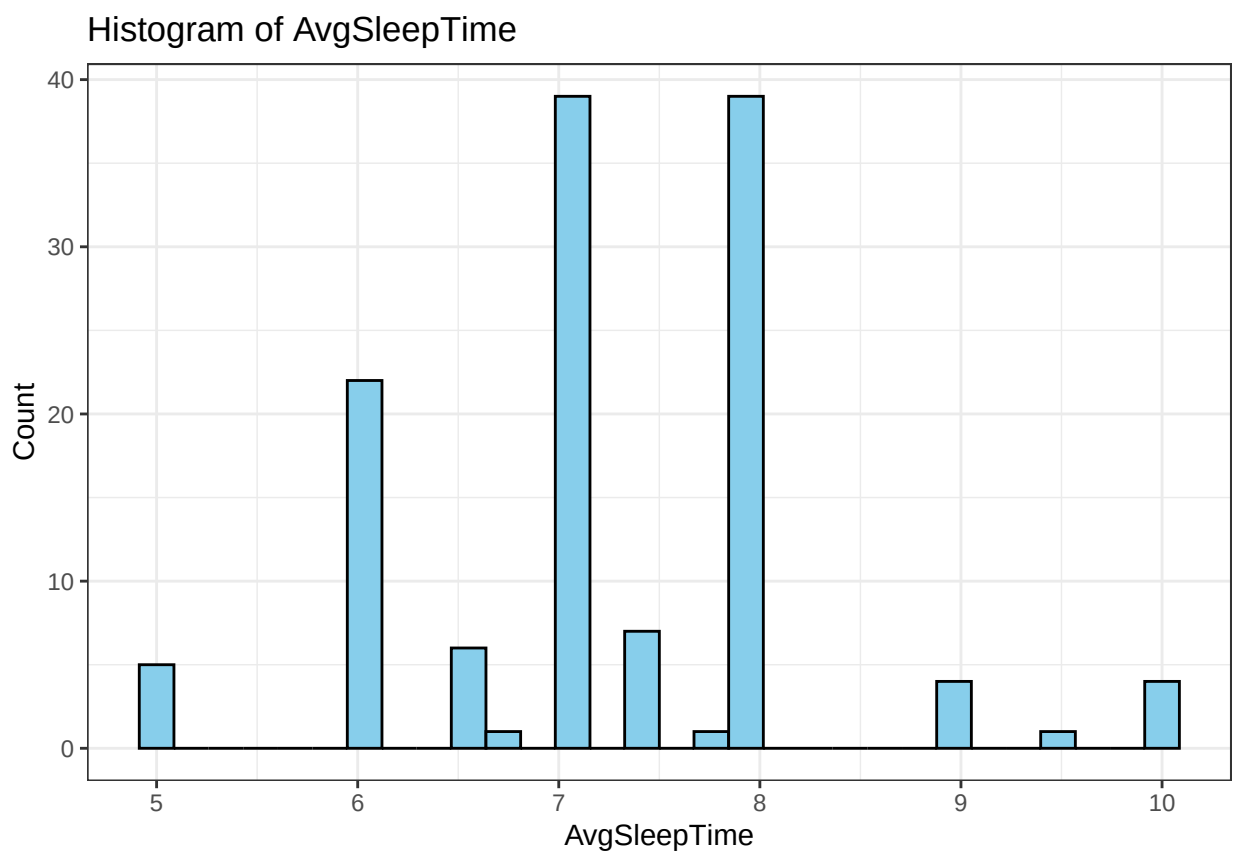


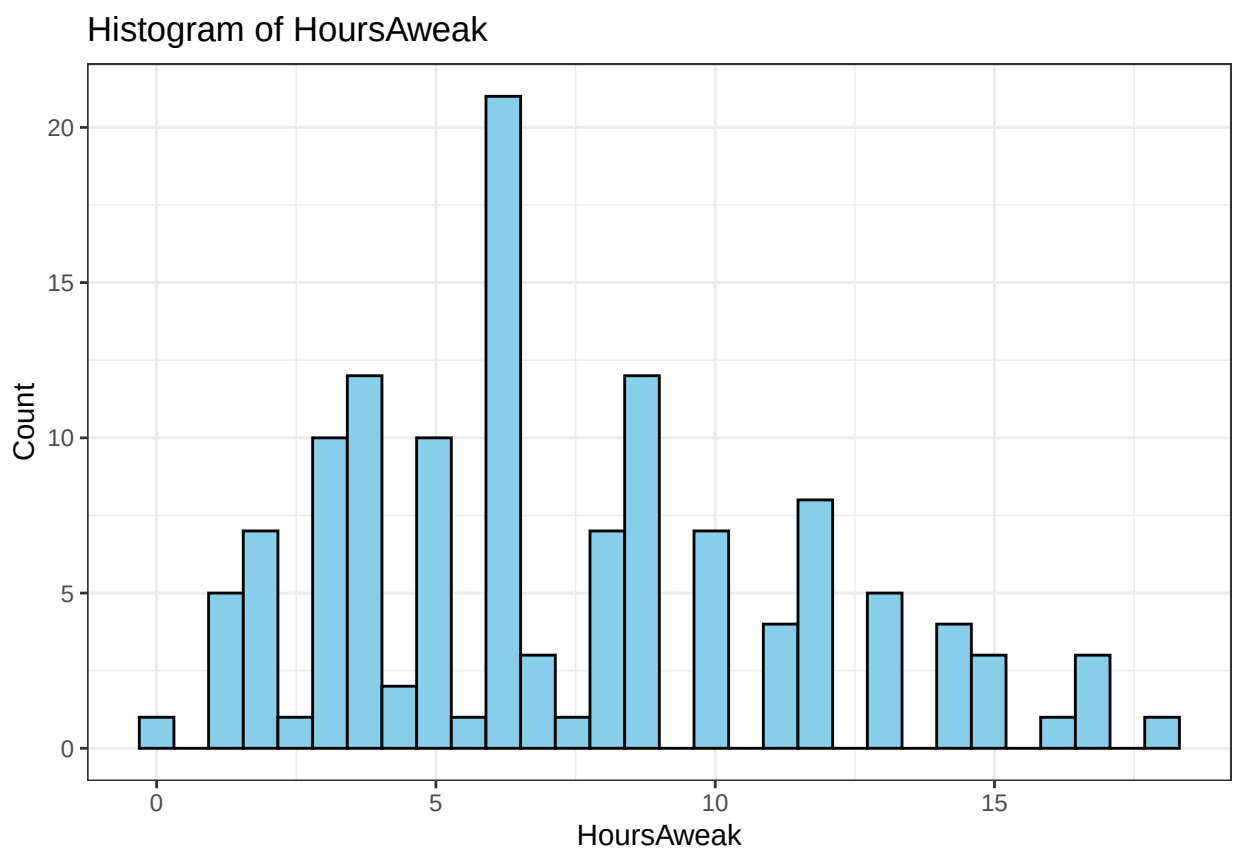


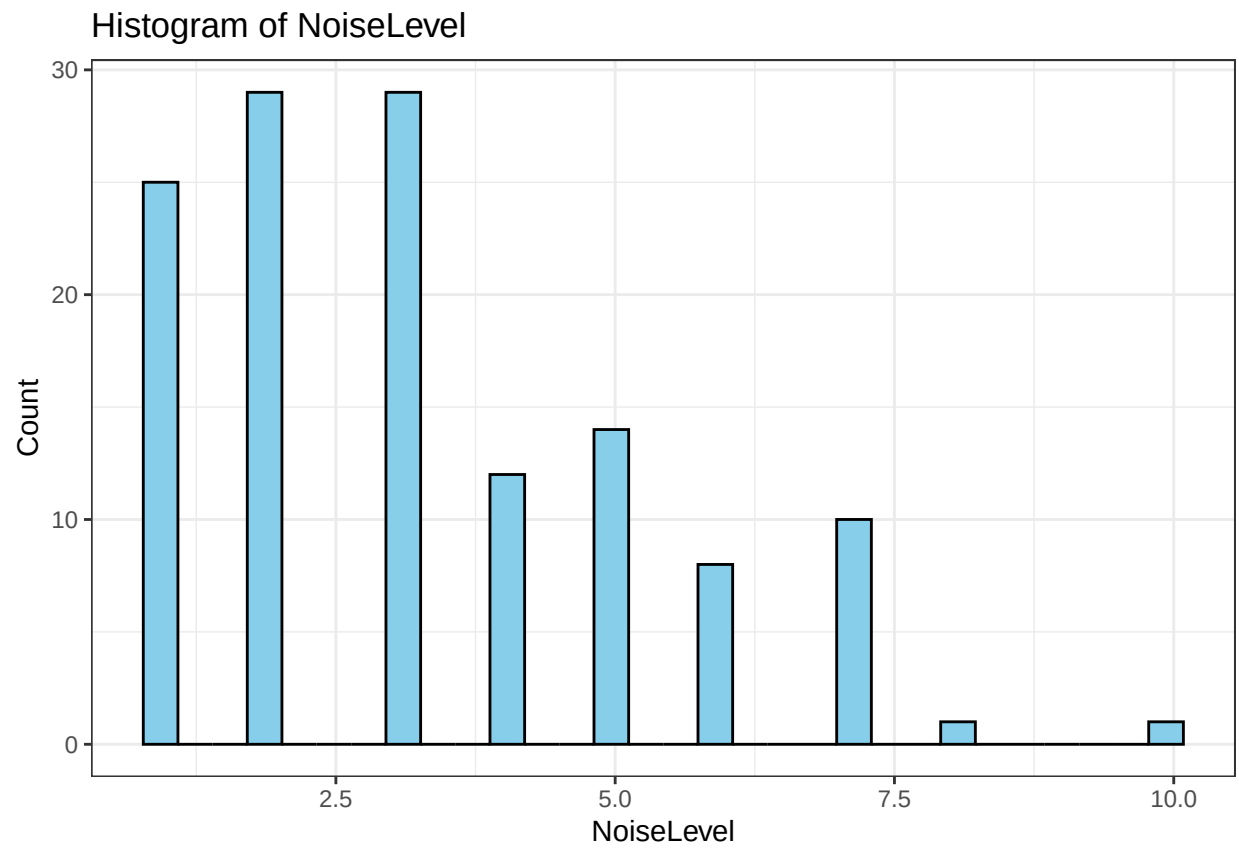




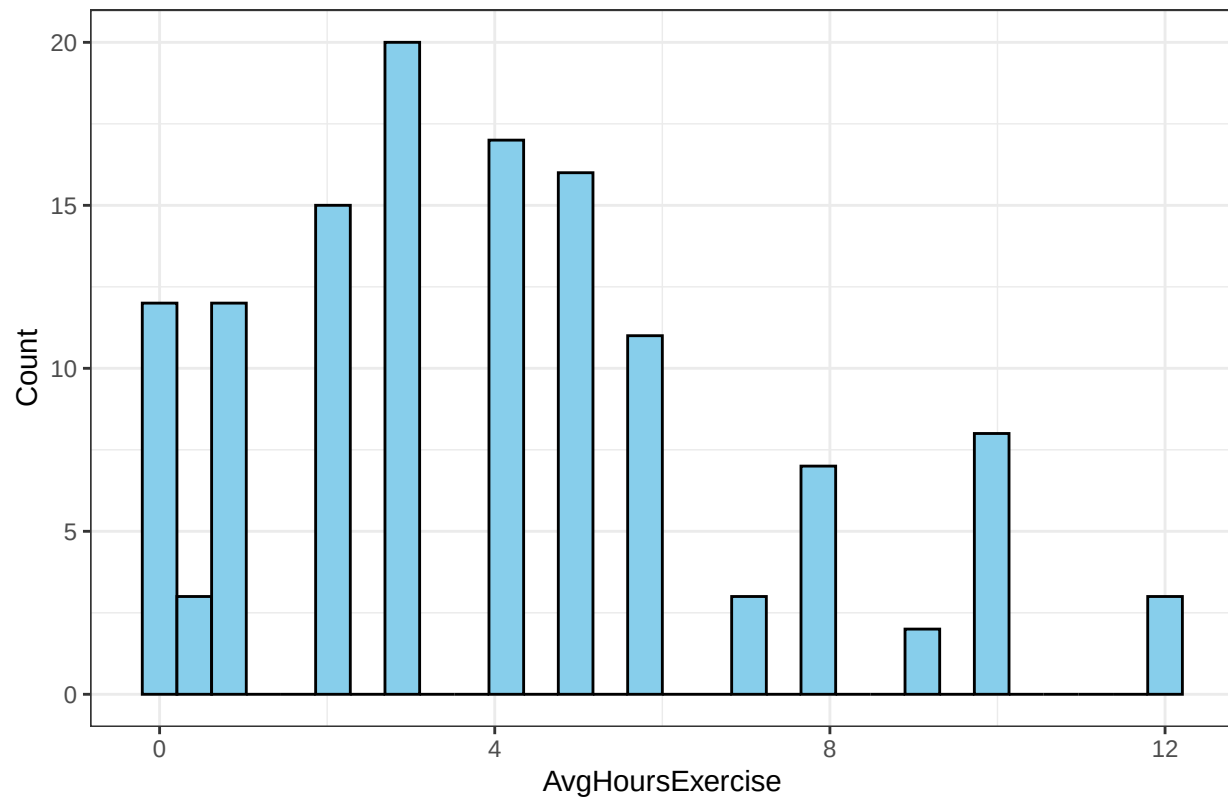








Histogram of AvgHoursExercise



```
##
## Call:
## lm(formula = avg_time ~ ., data = Survey_cleaned)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
	-119.584	-31.677	-1.699	23.259	158.041

```
##
## Coefficients:
```

	Estimate	Std. Error	t value
(Intercept)	316.7155	214.0247	1.480
ClassJunior	36.3028	19.3107	1.880
ClassSenior	5.9146	27.0350	0.219
ClassGraduate student	120.6718	55.3700	2.179
Age	3.0620	8.1732	0.375
AvgSleepTime	15.7785	8.3699	1.885
LastNightSleep	1.7912	5.3886	0.332
HoursAweek	0.5274	2.0285	0.260
DistractionYes	28.4336	20.0834	1.416
TemperatureWarm	25.1289	64.2955	0.391
TemperatureNeutral	46.3851	60.6552	0.765
TemperatureCold	65.9307	63.1520	1.044
GamerYes	15.4718	18.0549	0.857
CaffeinIntakeYes	40.1718	17.9718	2.235
VisualAcuityExcellent	-58.0343	29.0626	-1.997
VisualAcuityGood	-79.1387	29.9986	-2.638

## VisualAcuityPoor	-72.0108	50.5460	-1.425
## VisualAcuityVery Poor	-146.9141	97.1368	-1.512
## PrimarilyHandLeft hand	-54.6740	44.0925	-1.240
## PrimarilyHandAmbidextrous (both hands equally)	34.9272	40.1681	0.870
## UsePrimaryHandNo	65.8135	40.6695	1.618
## CautiousVery cautious	3.0895	45.3541	0.068
## CautiousModerately cautious	-6.5308	46.0642	-0.142
## CautiousSlightly cautious	9.0990	48.4746	0.188
## CautiousNot cautious at all	38.3024	64.5652	0.593
## InputDeviceMouse	-45.1659	78.2158	-0.577
## InputDeviceKeyboard	77.1231	96.5755	0.799
## InputDeviceTrackpad	37.3924	77.5345	0.482
## InputDeviceTouch screen	18.3343	85.9597	0.213
## RefreshRateMid-Range	30.3461	24.3793	1.245
## RefreshRateStandard	59.9600	22.9308	2.615
## DeviceiPad(Tablet	-42.5427	64.8587	-0.656
## DeviceiPhone(Smartphone	81.0134	83.1190	0.975
## DeviceLaptop	36.2946	24.0861	1.507
## WiFiStable	-96.2096	58.0745	-1.657
## Fatigue_3Moderately Fatigue	54.4543	26.0192	2.093
## Fatigue_3Low Fatigue	48.2829	26.9053	1.795
## Stress_3Moderately Stress	14.0792	20.5947	0.684
## Stress_3Low Stress	-29.3721	21.5548	-1.363
## Noise_LevelModerately Noise	-35.7234	71.8799	-0.497
## Noise_LevelLow Noise	-19.1215	71.4386	-0.268
## Game_FrequencyLow Frequency	21.8759	15.1069	1.448
## Sports_FrequencyLow Frequency	17.6599	18.2598	0.967
## Shooting_LevelsIntermediate Levels	-7.1078	24.4294	-0.291
## Shooting_LevelsLow Levels	21.0557	22.7596	0.925
## SystemLinux	39.9766	67.0464	0.596
## SystemWindows	-0.8867	19.2729	-0.046
## logAvgHoursExercise	3.1802	12.8470	0.248
##	Pr(> t)		
## (Intercept)	0.1428		
## ClassJunior	0.0637 .		
## ClassSenior	0.8274		
## ClassGraduate student	0.0322 *		
## Age	0.7089		
## AvgSleepTime	0.0630 .		
## LastNightSleep	0.7404		
## HoursAweak	0.7955		
## DistractionYes	0.1607		
## TemperatureWarm	0.6969		
## TemperatureNeutral	0.4467		
## TemperatureCold	0.2996		
## GamerYes	0.3940		
## CaffeinIntakeYes	0.0282 *		
## VisualAcuityExcellent	0.0492 *		
## VisualAcuityGood	0.0100 **		
## VisualAcuityPoor	0.1581		
## VisualAcuityVery Poor	0.1343		
## PrimarilyHandLeft hand	0.2186		
## PrimarilyHandAmbidextrous (both hands equally)	0.3871		
## UsePrimaryHandNo	0.1095		

```

## CautiousVery cautious 0.9459
## CautiousModerately cautious 0.8876
## CautiousSlightly cautious 0.8516
## CautiousNot cautious at all 0.5547
## InputDeviceMouse 0.5652
## InputDeviceKeyboard 0.4269
## InputDeviceTrackpad 0.6309
## InputDeviceTouch screen 0.8316
## RefreshRateMid-Range 0.2168
## RefreshRateStandard 0.0106 *
## DeviceiPad(Tablet 0.5137
## DeviceiPhone(Smartphone 0.3326
## DeviceLaptop 0.1357
## WiFiStable 0.1015
## Fatigue_3Moderately Fatigue 0.0395 *
## Fatigue_3Low Fatigue 0.0765 .
## Stress_3Moderately Stress 0.4962
## Stress_3Low Stress 0.1768
## Noise_LevelModerately Noise 0.6205
## Noise_LevelLow Noise 0.7896
## Game_FrequencyLow Frequency 0.1515
## Sports_FrequencyLow Frequency 0.3364
## Shooting_LevelsIntermediate Levels 0.7718
## Shooting_LevelsLow Levels 0.3576
## SystemLinux 0.5527
## SystemWindows 0.9634
## logAvgHoursExercise 0.8051
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 66.77 on 81 degrees of freedom
## Multiple R-squared:  0.6914, Adjusted R-squared:  0.5123
## F-statistic: 3.861 on 47 and 81 DF,  p-value: 4.833e-08

## Start:  AIC=1119.88
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Temperature + Gamer + CaffeinIntake + VisualAcuity +
##   PrimarilyHand + UsePrimaryHand + Cautious + InputDevice +
##   RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Noise_Level +
##   Game_Frequency + Sports_Frequency + Shooting_Levels + System +
##   logAvgHoursExercise
##
##           Df Sum of Sq  RSS    AIC
## - Cautious      4      6186 367259 1114.1
## - System        2       1980 363052 1116.6
## - Noise_Level    2       5074 366147 1117.7
## - Temperature    3     10845 371918 1117.7
## - logAvgHoursExercise 1        273 361346 1118.0
## - HoursAweak     1        301 361374 1118.0
## - LastNightSleep 1        493 361565 1118.0
## - Age            1        626 361699 1118.1
## - Gamer          1       3273 364346 1119.0
## - Sports_Frequency 1       4170 365243 1119.4
## <none>              361073 1119.9

```

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## - PrimarilyHand      2      11526 372599 1119.9
## - Shooting_Levels    2      11896 372969 1120.1
## - Distraction        1       8935 370008 1121.0
## - Game_Frequency     1       9347 370420 1121.2
## - Device             3      22755 383828 1121.8
## - UsePrimaryHand     1      11674 372746 1122.0
## - WiFi              1      12234 373307 1122.2
## - Fatigue_3          2      19740 380813 1122.7
## - AvgSleepTime       1      15842 376915 1123.4
## - Stress_3           2      23328 384401 1124.0
## - VisualAcuity       4      37145 398218 1124.5
## - CaffeinIntake      1      22272 383345 1125.6
## - Class              3      40144 401217 1127.5
## - RefreshRate        2      34921 395994 1127.8
## - InputDevice        4      73291 434364 1135.7
##
## Step: AIC=1114.07
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Temperature + Gamer + CaffeinIntake + VisualAcuity +
##   PrimarilyHand + UsePrimaryHand + InputDevice + RefreshRate +
##   Device + WiFi + Fatigue_3 + Stress_3 + Noise_Level + Game_Frequency +
##   Sports_Frequency + Shooting_Levels + System + logAvgHoursExercise
##
##           Df Sum of Sq   RSS   AIC
## - System      2      3458 370717 1111.3
## - Temperature  3     11208 378467 1112.0
## - logAvgHoursExercise 1         0 367259 1112.1
## - Age          1       450 367708 1112.2
## - LastNightSleep 1       519 367778 1112.2
## - HoursAweak   1       611 367869 1112.3
## - Noise_Level  2     8204 375463 1112.9
## - Gamer        1     3617 370875 1113.3
## - Sports_Frequency 1     4109 371368 1113.5
## - PrimarilyHand 2    10345 377604 1113.7
## <none>                367259 1114.1
## - Shooting_Levels  2    14422 381681 1115.0
## - Distraction     1    10066 377324 1115.6
## - Device          3    22527 389786 1115.8
## - Game_Frequency  1    10895 378153 1115.8
## - WiFi           1    12477 379736 1116.4
## - UsePrimaryHand  1    15475 382733 1117.4
## - Fatigue_3       2    23869 391128 1118.2
## - Stress_3        2    23955 391213 1118.2
## - VisualAcuity    4    39758 407017 1119.3
## - CaffeinIntake   1    22391 389650 1119.7
## + Cautious        4       6186 361073 1119.9
## - AvgSleepTime    1    26198 393457 1121.0
## - Class           3    40028 407286 1121.4
## - RefreshRate     2    36741 404000 1122.4
## - InputDevice     4    74330 441589 1129.8
##
## Step: AIC=1111.28
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Temperature + Gamer + CaffeinIntake + VisualAcuity +

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##      PrimarilyHand + UsePrimaryHand + InputDevice + RefreshRate +
##      Device + WiFi + Fatigue_3 + Stress_3 + Noise_Level + Game_Frequency +
##      Sports_Frequency + Shooting_Levels + logAvgHoursExercise
##
##      Df Sum of Sq    RSS    AIC
## - logAvgHoursExercise 1      18 370734 1109.3
## - HoursAweak          1      91 370807 1109.3
## - Temperature         3    11934 382651 1109.4
## - Noise_Level         2     6813 377529 1109.6
## - Age                 1     1077 371794 1109.7
## - LastNightSleep      1     1080 371797 1109.7
## - Gamer               1     4380 375097 1110.8
## - PrimarilyHand       2    10668 381384 1110.9
## - Sports_Frequency    1     5524 376240 1111.2
## <none>                370717 1111.3
## - Distraction         1     9422 380139 1112.5
## - Shooting_Levels     2    15720 386437 1112.6
## - Game_Frequency      1    10570 381286 1112.9
## - Device              3    23286 394002 1113.1
## - WiFi                1    13732 384449 1114.0
## + System              2     3458 367259 1114.1
## - Fatigue_3           2    21150 391867 1114.4
## - UsePrimaryHand      1    16142 386858 1114.8
## - Stress_3            2    26900 397616 1116.3
## + Cautious            4     7664 363052 1116.6
## - VisualAcuity        4    40603 411320 1116.7
## - CaffeinIntake       1    23349 394066 1117.2
## - AvgSleepTime        1    23564 394280 1117.2
## - Class               3    37904 408621 1117.8
## - RefreshRate         2    33438 404154 1118.4
## - InputDevice         4   105407 476123 1135.6
##
## Step:  AIC=1109.28
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##      Distraction + Temperature + Gamer + CaffeinIntake + VisualAcuity +
##      PrimarilyHand + UsePrimaryHand + InputDevice + RefreshRate +
##      Device + WiFi + Fatigue_3 + Stress_3 + Noise_Level + Game_Frequency +
##      Sports_Frequency + Shooting_Levels
##
##      Df Sum of Sq    RSS    AIC
## - HoursAweak          1      87 370821 1107.3
## - Temperature         3    11961 382695 1107.4
## - Noise_Level         2     6820 377555 1107.6
## - Age                 1     1060 371794 1107.7
## - LastNightSleep      1     1064 371798 1107.7
## - Gamer               1     4366 375100 1108.8
## - PrimarilyHand       2    10827 381561 1109.0
## <none>                370734 1109.3
## - Sports_Frequency    1     5919 376653 1109.3
## - Distraction         1     9495 380229 1110.5
## - Shooting_Levels     2    16442 387176 1110.9
## - Game_Frequency      1    10647 381381 1110.9
## - Device              3    23274 394008 1111.1
## + logAvgHoursExercise 1      18 370717 1111.3

```

```

## - WiFi 1 13743 384477 1112.0
## + System 2 3475 367259 1112.1
## - Fatigue_3 2 21828 392562 1112.7
## - UsePrimaryHand 1 16130 386865 1112.8
## - Stress_3 2 26889 397623 1114.3
## + Cautious 4 7260 363474 1114.7
## - VisualAcuity 4 40924 411658 1114.8
## - AvgSleepTime 1 23825 394559 1115.3
## - CaffeinIntake 1 24115 394849 1115.4
## - Class 3 37930 408664 1115.8
## - RefreshRate 2 34559 405293 1116.8
## - InputDevice 4 105750 476484 1133.7
##
## Step: AIC=1107.31
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + Distraction +
## Temperature + Gamer + CaffeinIntake + VisualAcuity + PrimarilyHand +
## UsePrimaryHand + InputDevice + RefreshRate + Device + WiFi +
## Fatigue_3 + Stress_3 + Noise_Level + Game_Frequency + Sports_Frequency +
## Shooting_Levels
##
## Df Sum of Sq RSS AIC
## - Temperature 3 11970 382791 1105.4
## - Noise_Level 2 6813 377635 1105.7
## - LastNightSleep 1 1066 371888 1105.7
## - Age 1 1105 371926 1105.7
## - Gamer 1 4474 375295 1106.9
## <none> 370821 1107.3
## - Sports_Frequency 1 6209 377030 1107.5
## - PrimarilyHand 2 12830 383651 1107.7
## - Distraction 1 9417 380238 1108.5
## - Game_Frequency 1 10560 381381 1108.9
## - Shooting_Levels 2 16837 387659 1109.0
## - Device 3 23213 394034 1109.2
## + HoursAweak 1 87 370734 1109.3
## + logAvgHoursExercise 1 14 370807 1109.3
## - WiFi 1 13749 384571 1110.0
## + System 2 2952 367870 1110.3
## - Fatigue_3 2 23362 394183 1111.2
## - UsePrimaryHand 1 17655 388477 1111.3
## - Stress_3 2 27142 397963 1112.4
## + Cautious 4 7320 363501 1112.7
## - VisualAcuity 4 41189 412010 1112.9
## - AvgSleepTime 1 23826 394648 1113.3
## - CaffeinIntake 1 24468 395289 1113.6
## - Class 3 38459 409280 1114.0
## - RefreshRate 2 34613 405434 1114.8
## - InputDevice 4 107672 478493 1132.2
##
## Step: AIC=1105.41
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + Distraction +
## Gamer + CaffeinIntake + VisualAcuity + PrimarilyHand + UsePrimaryHand +
## InputDevice + RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 +
## Noise_Level + Game_Frequency + Sports_Frequency + Shooting_Levels
##

```

```

##          Df Sum of Sq    RSS    AIC
## - Noise_Level      2      4281 387073 1102.8
## - LastNightSleep    1       284 383075 1103.5
## - Age               1      1279 384070 1103.8
## - Gamer             1      1827 384618 1104.0
## - Sports_Frequency  1      5120 387912 1105.1
## <none>                382791 1105.4
## - Shooting_Levels   2     12603 395394 1105.6
## - Distraction        1      8186 390978 1106.1
## - Game_Frequency    1      8210 391001 1106.2
## - PrimarilyHand     2     16490 399282 1106.8
## - Device            3     23846 406638 1107.2
## + Temperature       3     11970 370821 1107.3
## + HoursAweak        1        97 382695 1107.4
## + logAvgHoursExercise 1        40 382751 1107.4
## + System            2      3673 379119 1108.2
## - WiFi             1     15089 397880 1108.4
## - Stress_3         2     23613 406404 1109.1
## - VisualAcuity      4     36566 419357 1109.2
## - Fatigue_3        2     28179 410971 1110.6
## - UsePrimaryHand    1     21981 404773 1110.6
## + Cautious         4       7923 374869 1110.7
## - AvgSleepTime      1     23256 406048 1111.0
## - Class            3     36590 419382 1111.2
## - CaffeinIntake     1     25129 407921 1111.6
## - RefreshRate       2     36072 418863 1113.0
## - InputDevice       4     100467 483258 1127.5
##
## Step:  AIC=1102.85
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + Distraction +
##           Gamer + CaffeinIntake + VisualAcuity + PrimarilyHand + UsePrimaryHand +
##           InputDevice + RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 +
##           Game_Frequency + Sports_Frequency + Shooting_Levels
##
##          Df Sum of Sq    RSS    AIC
## - LastNightSleep    1       383 387456 1101.0
## - Gamer             1     1318 388391 1101.3
## - Age              1     1614 388687 1101.4
## - Shooting_Levels   2     11193 398266 1102.5
## <none>                387073 1102.8
## - Distraction        1     6132 393205 1102.9
## - Sports_Frequency   1     6408 393481 1103.0
## - Game_Frequency    1     8205 395278 1103.5
## - PrimarilyHand     2    16349 403422 1104.2
## + HoursAweak        1        74 386999 1104.8
## + logAvgHoursExercise 1         1 387071 1104.8
## + Noise_Level       2      4281 382791 1105.4
## - WiFi             1    14680 401753 1105.7
## + Temperature       3      9438 377635 1105.7
## + System            2      2576 384497 1106.0
## - Stress_3         2     23485 410557 1106.4
## - Device            3     31250 418322 1106.9
## - VisualAcuity      4     38848 425921 1107.2
## + Cautious         4      9580 377493 1107.6

```

```

## - Fatigue_3          2      27266 414338 1107.6
## - Class              3      34189 421262 1107.8
## - AvgSleepTime       1      23007 410080 1108.3
## - CaffeinIntake      1      23850 410922 1108.6
## - UsePrimaryHand     1      24556 411629 1108.8
## - RefreshRate        2      35695 422768 1110.2
## - InputDevice        4      98624 485696 1124.1
##
## Step: AIC=1100.97
## avg_time ~ Class + Age + AvgSleepTime + Distraction + Gamer +
##   CaffeinIntake + VisualAcuity + PrimarilyHand + UsePrimaryHand +
##   InputDevice + RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 +
##   Game_Frequency + Sports_Frequency + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - Gamer          1      1334 388790 1099.4
## - Age            1      1828 389284 1099.6
## - Shooting_Levels 2     11178 398633 1100.6
## - Distraction    1      5951 393406 1100.9
## <none>                387456 1101.0
## - Sports_Frequency 1      6316 393772 1101.1
## - Game_Frequency  1      8139 395594 1101.7
## - PrimarilyHand   2     16257 403712 1102.3
## + LastNightSleep  1       383 387073 1102.8
## + HoursAweak      1        76 387380 1103.0
## + logAvgHoursExercise 1        17 387438 1103.0
## + Noise_Level     2      4380 383075 1103.5
## - WiFi            1     14370 401826 1103.7
## + Temperature     3      8701 378754 1104.0
## + System           2      2691 384764 1104.1
## - Stress_3        2     23111 410566 1104.5
## - Device          3     31664 419119 1105.1
## - VisualAcuity    4     38508 425964 1105.2
## + Cautious        4      9643 377812 1105.7
## - Class           3     34943 422398 1106.1
## - CaffeinIntake   1     23961 411416 1106.7
## - Fatigue_3       2     30543 417998 1106.8
## - UsePrimaryHand  1     24445 411900 1106.9
## - RefreshRate     2     35509 422965 1108.3
## - AvgSleepTime    1     29291 416746 1108.4
## - InputDevice     4    100671 488127 1122.8
##
## Step: AIC=1099.42
## avg_time ~ Class + Age + AvgSleepTime + Distraction + CaffeinIntake +
##   VisualAcuity + PrimarilyHand + UsePrimaryHand + InputDevice +
##   RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Game_Frequency +
##   Sports_Frequency + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - Age          1      2399 391188 1098.2
## - Shooting_Levels 2     10171 398960 1098.8
## - Distraction   1      5525 394315 1099.2
## - Sports_Frequency 1      5816 394606 1099.3
## <none>                388790 1099.4

```

```

## - Game_Frequency      1      6920 395710 1099.7
## - PrimarilyHand       2     15844 404634 1100.6
## + Gamer               1      1334 387456 1101.0
## + LastNightSleep      1       399 388391 1101.3
## + HoursAweak          1       123 388667 1101.4
## + logAvgHoursExercise  1        55 388734 1101.4
## - WiFi                1     13216 402006 1101.7
## + Noise_Level         2      3866 384924 1102.1
## + System              2      2804 385985 1102.5
## - Stress_3            2     22081 410871 1102.5
## + Temperature         3      6969 381821 1103.1
## - VisualAcuity        4     37179 425969 1103.2
## - Device              3     31474 420264 1103.5
## + Cautious            4      9881 378909 1104.1
## - Class               3     33671 422461 1104.1
## - CaffeinIntake       1     22733 411522 1104.8
## - Fatigue_3           2     29498 418288 1104.8
## - UsePrimaryHand      1     23127 411916 1104.9
## - AvgSleepTime        1     28824 417614 1106.6
## - RefreshRate         2     35662 424452 1106.7
## - InputDevice         4    100536 489325 1121.1
##
## Step: AIC=1098.21
## avg_time ~ Class + AvgSleepTime + Distraction + CaffeinIntake +
##      VisualAcuity + PrimarilyHand + UsePrimaryHand + InputDevice +
##      RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Game_Frequency +
##      Sports_Frequency + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - Shooting_Levels      2      9379 400568 1097.3
## - Distraction           1      5870 397058 1098.1
## <none>                  391188 1098.2
## - Sports_Frequency      1      7300 398489 1098.6
## - Game_Frequency        1      7549 398738 1098.7
## + Age                   1      2399 388790 1099.4
## + Gamer                 1      1905 389284 1099.6
## - PrimarilyHand         2     17520 408708 1099.9
## + LastNightSleep        1       661 390527 1100.0
## - WiFi                  1     11941 403130 1100.1
## + logAvgHoursExercise   1       275 390913 1100.1
## + HoursAweak            1       211 390977 1100.1
## + Noise_Level           2      4203 386986 1100.8
## + System                2      3594 387594 1101.0
## - Stress_3              2     22189 413377 1101.3
## - VisualAcuity          4     35439 426628 1101.4
## + Temperature           3      6681 384507 1102.0
## - Device                3     32199 423387 1102.4
## + Cautious              4     10431 380757 1102.7
## - Fatigue_3             2     30846 422034 1104.0
## - UsePrimaryHand        1     25826 417014 1104.5
## - CaffeinIntake         1     25935 417123 1104.5
## - AvgSleepTime          1     26988 418176 1104.8
## - RefreshRate           2     35935 427124 1105.5
## - Class                 3     80062 471250 1116.2

```

```

## - InputDevice          4      99795 490984 1119.5
##
## Step: AIC=1097.27
## avg_time ~ Class + AvgSleepTime + Distraction + CaffeinIntake +
##      VisualAcuity + PrimarilyHand + UsePrimaryHand + InputDevice +
##      RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Game_Frequency +
##      Sports_Frequency
##
##              Df Sum of Sq    RSS    AIC
## - Distraction      1      5333 405900 1097.0
## <none>                      400568 1097.3
## - WiFi              1      8805 409372 1098.1
## - PrimarilyHand     2     15495 416062 1098.2
## + Shooting_Levels   2      9379 391188 1098.2
## - Sports_Frequency  1     10001 410569 1098.5
## + Age               1      1607 398960 1098.8
## + logAvgHoursExercise 1      1285 399283 1098.8
## + Gamer             1       605 399962 1099.1
## + LastNightSleep    1       592 399976 1099.1
## + HoursAweak        1       476 400092 1099.1
## - Game_Frequency    1     12442 413010 1099.2
## + System            2       4113 396455 1099.9
## + Noise_Level       2       2983 397584 1100.3
## - Stress_3          2     24562 425129 1100.9
## + Cautious          4     11847 388721 1101.4
## + Temperature       3       4947 395620 1101.7
## - VisualAcuity      4     40553 441121 1101.7
## - UsePrimaryHand    1     20795 421363 1101.8
## - Device            3     37731 438298 1102.9
## - CaffeinIntake     1     25512 426080 1103.2
## - AvgSleepTime      1     25819 426386 1103.3
## - Fatigue_3         2     39403 439970 1105.4
## - RefreshRate       2     40891 441458 1105.8
## - Class             3     94721 495289 1118.7
## - InputDevice       4     111509 512077 1121.0
##
## Step: AIC=1096.97
## avg_time ~ Class + AvgSleepTime + CaffeinIntake + VisualAcuity +
##      PrimarilyHand + UsePrimaryHand + InputDevice + RefreshRate +
##      Device + WiFi + Fatigue_3 + Stress_3 + Game_Frequency + Sports_Frequency
##
##              Df Sum of Sq    RSS    AIC
## <none>                      405900 1097.0
## - PrimarilyHand     2     13237 419137 1097.1
## + Distraction       1      5333 400568 1097.3
## - WiFi              1      8406 414306 1097.6
## - Sports_Frequency  1      8986 414886 1097.8
## + Shooting_Levels   2     8842 397058 1098.1
## + Age               1      1943 403957 1098.3
## + logAvgHoursExercise 1      1905 403995 1098.4
## - Game_Frequency    1     11911 417811 1098.7
## + LastNightSleep    1       428 405473 1098.8
## + HoursAweak        1       399 405501 1098.8
## + Gamer             1       398 405503 1098.8

```

```

## + System                2      3809 402091 1099.8
## + Noise_Level           2      1132 404768 1100.6
## + Cautious              4     12107 393793 1101.1
## + Temperature          3      4990 400910 1101.4
## - UsePrimaryHand       1     20798 426698 1101.4
## - Stress_3             2     28495 434395 1101.7
## - VisualAcuity         4     43516 449416 1102.1
## - AvgSleepTime        1     24693 430593 1102.6
## - Device               3     38320 444220 1102.6
## - CaffeinIntake        1     33563 439463 1105.2
## - Fatigue_3            2     41583 447483 1105.5
## - RefreshRate          2     42714 448614 1105.9
## - Class                 3     93752 499652 1117.8
## - InputDevice          4    115093 520993 1121.2

##
## Call:
## lm(formula = avg_time ~ Class + AvgSleepTime + CaffeinIntake +
##     VisualAcuity + PrimarilyHand + UsePrimaryHand + InputDevice +
##     RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Game_Frequency +
##     Sports_Frequency, data = Survey_cleaned)
##
## Coefficients:
##                (Intercept)
##                438.055
##                ClassJunior
##                38.454
##                ClassSenior
##                15.073
##      ClassGraduate student
##                129.289
##                AvgSleepTime
##                15.025
##                CaffeinIntakeYes
##                43.014
##      VisualAcuityExcellent
##                -69.927
##                VisualAcuityGood
##                -83.490
##                VisualAcuityPoor
##                -73.772
##      VisualAcuityVery Poor
##                -116.392
##      PrimarilyHandLeft hand
##                -51.238
## PrimarilyHandAmbidextrous (both hands equally)
##                31.844
##                UsePrimaryHandNo
##                74.840
##                InputDeviceMouse
##                -51.908
##                InputDeviceKeyboard
##                53.504
##                InputDeviceTrackpad

```

```

##                25.861
##      InputDeviceTouch screen
##                -20.490
##      RefreshRateMid-Range
##                27.233
##      RefreshRateStandard
##                57.230
##      DeviceiPad(Tablet
##                -24.929
##      DeviceiPhone(Smartphone
##                142.767
##      DeviceLaptop
##                36.504
##      WiFiStable
##                -71.611
##      Fatigue_3Moderately Fatigue
##                63.388
##      Fatigue_3Low Fatigue
##                56.646
##      Stress_3Moderately Stress
##                7.428
##      Stress_3Low Stress
##                -32.981
##      Game_FrequencyLow Frequency
##                21.879
##      Sports_FrequencyLow Frequency
##                20.661

## Start:  AIC=1257.15
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Temperature + Gamer + CaffeinIntake + VisualAcuity +
##   PrimarilyHand + UsePrimaryHand + Cautious + InputDevice +
##   RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Noise_Level +
##   Game_Frequency + Sports_Frequency + Shooting_Levels + System +
##   logAvgHoursExercise
##
##      Df Sum of Sq  RSS    AIC
## - Cautious      4      6186 367259 1239.9
## - Temperature    3     10845 371918 1246.4
## - System          2      1980 363052 1248.1
## - Noise_Level     2      5074 366147 1249.2
## - VisualAcuity    4      37145 398218 1250.3
## - Device          3      22755 383828 1250.5
## - PrimarilyHand   2      11526 372599 1251.5
## - Shooting_Levels  2      11896 372969 1251.6
## - logAvgHoursExercise 1        273 361346 1252.4
## - HoursAweak      1        301 361374 1252.4
## - LastNightSleep   1        493 361565 1252.5
## - Age              1        626 361699 1252.5
## - Gamer            1       3273 364346 1253.5
## - Sports_Frequency  1       4170 365243 1253.8
## - Fatigue_3        2      19740 380813 1254.3
## - Distraction      1       8935 370008 1255.4
## - Stress_3         2      23328 384401 1255.5

```

```

## - Game_Frequency      1      9347 370420 1255.6
## - Class                3      40144 401217 1256.2
## - UsePrimaryHand       1      11674 372746 1256.4
## - WiFi                 1      12234 373307 1256.6
## <none>                  361073 1257.2
## - AvgSleepTime         1      15842 376915 1257.8
## - RefreshRate          2      34921 395994 1259.3
## - CaffeinIntake        1      22272 383345 1260.0
## - InputDevice          4      73291 434364 1261.5
##
## Step: AIC=1239.9
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Temperature + Gamer + CaffeinIntake + VisualAcuity +
##   PrimarilyHand + UsePrimaryHand + InputDevice + RefreshRate +
##   Device + WiFi + Fatigue_3 + Stress_3 + Noise_Level + Game_Frequency +
##   Sports_Frequency + Shooting_Levels + System + logAvgHoursExercise
##
##           Df Sum of Sq   RSS   AIC
## - Temperature      3    11208 378467 1229.2
## - System            2     3458 370717 1231.4
## - Device            3    22527 389786 1233.0
## - Noise_Level       2     8204 375463 1233.0
## - VisualAcuity      4    39758 407017 1233.7
## - PrimarilyHand     2    10345 377604 1233.8
## - logAvgHoursExercise 1         0 367259 1235.0
## - Shooting_Levels   2    14422 381681 1235.2
## - Age               1     450 367708 1235.2
## - LastNightSleep    1     519 367778 1235.2
## - HoursAweak        1     611 367869 1235.2
## - Gamer             1     3617 370875 1236.3
## - Sports_Frequency  1     4109 371368 1236.5
## - Fatigue_3         2    23869 391128 1238.3
## - Stress_3          2    23955 391213 1238.3
## - Distraction       1    10066 377324 1238.5
## - Class             3    40028 407286 1238.7
## - Game_Frequency    1    10895 378153 1238.8
## - WiFi              1    12477 379736 1239.3
## <none>              367259 1239.9
## - UsePrimaryHand    1    15475 382733 1240.4
## - RefreshRate       2    36741 404000 1242.5
## - CaffeinIntake     1    22391 389650 1242.7
## - AvgSleepTime      1    26198 393457 1243.9
## - InputDevice       4    74330 441589 1244.2
## + Cautious          4     6186 361073 1257.2
##
## Step: AIC=1229.2
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Gamer + CaffeinIntake + VisualAcuity + PrimarilyHand +
##   UsePrimaryHand + InputDevice + RefreshRate + Device + WiFi +
##   Fatigue_3 + Stress_3 + Noise_Level + Game_Frequency + Sports_Frequency +
##   Shooting_Levels + System + logAvgHoursExercise
##
##           Df Sum of Sq   RSS   AIC
## - System            2     4184 382651 1220.9

```

```

## - VisualAcuity      4      35557 414024 1221.3
## - Noise_Level       2        5619 384085 1221.4
## - Device            3      23281 401748 1222.3
## - Shooting_Levels   2      10787 389254 1223.1
## - PrimarilyHand     2      13084 391551 1223.9
## - logAvgHoursExercise 1         4 378471 1224.3
## - LastNightSleep    1         78 378545 1224.4
## - Age               1         538 379005 1224.5
## - HoursAweak        1         650 379117 1224.6
## - Gamer             1        1425 379892 1224.8
## - Sports_Frequency  1        3391 381858 1225.5
## - Stress_3          2      20992 399459 1226.4
## - Class             3      38229 416696 1227.0
## - Game_Frequency    1        8495 386962 1227.2
## - Distraction        1        8793 387260 1227.3
## - Fatigue_3         2      27884 406351 1228.7
## - WiFi              1      13749 392215 1228.9
## <none>              378467 1229.2
## - UsePrimaryHand    1      19388 397855 1230.8
## - InputDevice        4      68310 446777 1231.2
## - CaffeinIntake      1      22626 401093 1231.8
## - RefreshRate        2      38413 416880 1232.0
## - AvgSleepTime       1      25529 403996 1232.8
## + Temperature        3      11208 367259 1239.9
## + Cautious           4        6549 371918 1246.4
##
## Step:  AIC=1220.9
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Gamer + CaffeinIntake + VisualAcuity + PrimarilyHand +
##   UsePrimaryHand + InputDevice + RefreshRate + Device + WiFi +
##   Fatigue_3 + Stress_3 + Noise_Level + Game_Frequency + Sports_Frequency +
##   Shooting_Levels + logAvgHoursExercise
##
##           Df Sum of Sq  RSS    AIC
## - Noise_Level      2      4347 386998 1212.6
## - VisualAcuity      4     36345 418996 1213.2
## - Device            3     23904 406555 1214.1
## - Shooting_Levels   2     11863 394514 1215.1
## - PrimarilyHand     2     13695 396346 1215.7
## - logAvgHoursExercise 1         44 382695 1216.0
## - HoursAweak        1         101 382751 1216.1
## - LastNightSleep    1         309 382960 1216.1
## - Age               1        1275 383925 1216.5
## - Gamer             1        1799 384450 1216.6
## - Sports_Frequency  1        4666 387317 1217.6
## - Class             3     36049 418700 1217.9
## - Game_Frequency    1        8180 390831 1218.8
## - Distraction        1        8254 390905 1218.8
## - Stress_3          2     23453 406104 1218.8
## - Fatigue_3         2     25061 407712 1219.4
## <none>              382651 1220.9
## - WiFi              1     15046 397697 1221.0
## - RefreshRate        2     34523 417174 1222.3
## - UsePrimaryHand    1     20006 402657 1222.6

```

```

## - AvgSleepTime          1      23174 405825 1223.6
## - CaffeinIntake         1      24173 406824 1223.9
## + System                2         4184 378467 1229.2
## - InputDevice           4      98241 480891 1230.9
## + Temperature           3      11934 370717 1231.4
## + Cautious              4         7997 374654 1237.6
##
## Step: AIC=1212.63
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Gamer + CaffeinIntake + VisualAcuity + PrimarilyHand +
##   UsePrimaryHand + InputDevice + RefreshRate + Device + WiFi +
##   Fatigue_3 + Stress_3 + Game_Frequency + Sports_Frequency +
##   Shooting_Levels + logAvgHoursExercise
##
##              Df Sum of Sq    RSS    AIC
## - VisualAcuity      4      38779 425777 1205.5
## - Shooting_Levels    2      10264 397262 1206.3
## - PrimarilyHand      2      13663 400661 1207.4
## - logAvgHoursExercise 1           1 386999 1207.8
## - HoursAweak         1          73 387071 1207.8
## - LastNightSleep     1          367 387365 1207.9
## - Device             3      31210 418208 1208.1
## - Gamer              1          1257 388255 1208.2
## - Age                1          1519 388517 1208.3
## - Class              3      33787 420785 1208.8
## - Sports_Frequency    1          5428 392426 1209.6
## - Distraction         1          6008 393006 1209.8
## - Stress_3           2      23074 410072 1210.4
## - Game_Frequency     1          8251 395250 1210.5
## - Fatigue_3          2      23851 410849 1210.6
## - WiFi               1      14667 401666 1212.6
## <none>                386998 1212.6
## - RefreshRate        2      34545 421543 1213.9
## - UsePrimaryHand     1      22667 409665 1215.1
## - AvgSleepTime       1      23015 410013 1215.2
## - CaffeinIntake       1      23215 410213 1215.3
## + Noise_Level        2          4347 382651 1220.9
## + System              2          2913 384085 1221.4
## - InputDevice         4      96745 483743 1222.0
## + Temperature         3          9469 377529 1224.0
## + Cautious            4          9588 377410 1228.8
##
## Step: AIC=1205.51
## avg_time ~ Class + Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Gamer + CaffeinIntake + PrimarilyHand + UsePrimaryHand +
##   InputDevice + RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 +
##   Game_Frequency + Sports_Frequency + Shooting_Levels + logAvgHoursExercise
##
##              Df Sum of Sq    RSS    AIC
## - Class              3      30472 456249 1199.8
## - Shooting_Levels    2      13920 439697 1199.9
## - Stress_3           2      14882 440659 1200.2
## - Gamer              1           5 425782 1200.7
## - logAvgHoursExercise 1          20 425797 1200.7

```

```

## - LastNightSleep      1      55 425832 1200.7
## - HoursAweak          1     126 425903 1200.7
## - Age                 1     547 426324 1200.8
## - PrimarilyHand       2    16925 442702 1200.8
## - Device              3    36091 461868 1201.4
## - Sports_Frequency    1     4645 430422 1202.0
## - Game_Frequency      1     6796 432573 1202.7
## - WiFi               1     8600 434376 1203.2
## - Distraction         1     8767 434544 1203.3
## - Fatigue_3           2    29208 454985 1204.3
## - CaffeinIntake       1    14298 440074 1204.9
## <none>                1     425777 1205.5
## - RefreshRate         2    36358 462134 1206.4
## - AvgSleepTime        1    24160 449937 1207.8
## - UsePrimaryHand      1    26149 451926 1208.3
## + VisualAcuity        4    38779 386998 1212.6
## + Noise_Level         2     6781 418996 1213.2
## + System              2     3121 422656 1214.3
## - InputDevice         4   111805 537582 1216.2
## + Temperature         3     5643 420134 1218.4
## + Cautious            4    12690 413087 1221.0
##
## Step:  AIC=1199.85
## avg_time ~ Age + AvgSleepTime + LastNightSleep + HoursAweak +
##      Distraction + Gamer + CaffeinIntake + PrimarilyHand + UsePrimaryHand +
##      InputDevice + RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 +
##      Game_Frequency + Sports_Frequency + Shooting_Levels + logAvgHoursExercise
##
##              Df Sum of Sq  RSS    AIC
## - PrimarilyHand      2    15200 471449 1194.4
## - Device              3    34447 490696 1194.7
## - logAvgHoursExercise 1      179 456428 1195.0
## - LastNightSleep     1      254 456503 1195.1
## - HoursAweak         1      318 456567 1195.1
## - Gamer              1      525 456774 1195.1
## - Sports_Frequency    1     3518 459767 1196.0
## - Shooting_Levels     2    21649 477898 1196.1
## - Game_Frequency      1     4527 460777 1196.3
## - Fatigue_3          2    23691 479940 1196.7
## - CaffeinIntake       1     7922 464172 1197.2
## - Distraction         1     9878 466127 1197.8
## - WiFi               1    10276 466525 1197.9
## - Stress_3           2    31047 487296 1198.6
## - RefreshRate         2    32563 488812 1199.0
## <none>                1     456249 1199.8
## - UsePrimaryHand      1    26461 482710 1202.3
## - AvgSleepTime        1    30040 486289 1203.2
## + Class              3    30472 425777 1205.5
## - InputDevice         4   102228 558477 1206.5
## - Age                1    42831 499080 1206.6
## + Noise_Level         2     4071 452178 1208.4
## + VisualAcuity        4    35464 420785 1208.8
## + System              2     1296 454953 1209.2
## + Temperature         3     5658 450591 1212.8

```

```

## + Cautious          4      11104 445145 1216.1
##
## Step: AIC=1194.36
## avg_time ~ Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Gamer + CaffeinIntake + UsePrimaryHand + InputDevice +
##   RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Game_Frequency +
##   Sports_Frequency + Shooting_Levels + logAvgHoursExercise
##
##           Df Sum of Sq    RSS    AIC
## - Device      3      24984 496434 1186.4
## - Shooting_Levels  2      17731 489180 1189.4
## - logAvgHoursExercise  1         35 471484 1189.5
## - LastNightSleep    1        138 471587 1189.5
## - Sports_Frequency   1        909 472359 1189.8
## - Gamer              1        921 472370 1189.8
## - HoursAweak         1       3828 475278 1190.5
## - Game_Frequency     1       4999 476449 1190.9
## - CaffeinIntake      1       6754 478203 1191.3
## - Distraction        1       6974 478423 1191.4
## - WiFi               1       9910 481360 1192.2
## - RefreshRate        2      30567 502017 1192.7
## - Fatigue_3          2      34343 505793 1193.7
## - UsePrimaryHand     1      17381 488830 1194.2
## <none>                      471449 1194.4
## - Stress_3          2      41305 512754 1195.5
## - AvgSleepTime      1      28267 499716 1197.0
## + PrimarilyHand      2      15200 456249 1199.8
## - Age                1      43111 514560 1200.8
## + Class               3      28747 442702 1200.8
## - InputDevice        4     110627 582077 1202.1
## + Noise_Level        2        5782 465667 1202.5
## + VisualAcuity       4      38136 433314 1202.9
## + System              2        2180 469269 1203.5
## + Temperature        3        7900 463549 1206.8
## + Cautious           4       13026 458423 1210.2
##
## Step: AIC=1186.44
## avg_time ~ Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Gamer + CaffeinIntake + UsePrimaryHand + InputDevice +
##   RefreshRate + WiFi + Fatigue_3 + Stress_3 + Game_Frequency +
##   Sports_Frequency + Shooting_Levels + logAvgHoursExercise
##
##           Df Sum of Sq    RSS    AIC
## - logAvgHoursExercise  1         17 496451 1181.6
## - LastNightSleep      1         482 496916 1181.7
## - Sports_Frequency     1         771 497205 1181.8
## - Gamer                1        1229 497663 1181.9
## - HoursAweak           1        2382 498815 1182.2
## - Game_Frequency       1        2441 498874 1182.2
## - Shooting_Levels      2       25184 521618 1183.1
## - CaffeinIntake        1        8974 505407 1183.9
## - Distraction          1       10184 506618 1184.2
## - Fatigue_3            2       30325 526759 1184.4
## - WiFi                 1       12878 509311 1184.9

```

```

## - UsePrimaryHand      1      15112 511545 1185.5
## <none>                  496434 1186.4
## - RefreshRate         2      43484 539917 1187.5
## - Stress_3            2      47521 543954 1188.5
## - AvgSleepTime        1      31235 527669 1189.5
## - Age                 1      38244 534678 1191.2
## + Class               3      28012 468421 1193.5
## + Device              3      24984 471449 1194.4
## + PrimarilyHand       2        5737 490696 1194.7
## + Noise_Level         2        5195 491238 1194.8
## + VisualAcuity        4      39298 457135 1195.2
## + System              2        1585 494848 1195.8
## + Temperature        3        6437 489997 1199.3
## - InputDevice         4     141996 638430 1199.5
## + Cautious            4       16082 480351 1201.6
##
## Step:  AIC=1181.59
## avg_time ~ Age + AvgSleepTime + LastNightSleep + HoursAweak +
##   Distraction + Gamer + CaffeinIntake + UsePrimaryHand + InputDevice +
##   RefreshRate + WiFi + Fatigue_3 + Stress_3 + Game_Frequency +
##   Sports_Frequency + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - LastNightSleep      1         540 496991 1176.9
## - Sports_Frequency     1         958 497408 1177.0
## - Gamer                1        1217 497668 1177.0
## - HoursAweak           1        2420 498871 1177.3
## - Game_Frequency       1        2427 498877 1177.3
## - Shooting_Levels      2       26592 523043 1178.6
## - CaffeinIntake        1        9107 505558 1179.1
## - Distraction          1       10562 507013 1179.4
## - Fatigue_3            2       31573 528023 1179.8
## - WiFi                 1       12865 509316 1180.0
## - UsePrimaryHand       1       15160 511611 1180.6
## <none>                  496451 1181.6
## - RefreshRate         2       44380 540831 1182.9
## - Stress_3            2       47944 544395 1183.8
## - AvgSleepTime        1       31500 527951 1184.7
## + logAvgHoursExercise  1          17 496434 1186.4
## - Age                 1       40034 536485 1186.7
## + Class               3       27886 468565 1188.7
## + Device              3       24967 471484 1189.5
## + PrimarilyHand       2        5594 490856 1189.8
## + Noise_Level         2        5212 491238 1189.9
## + VisualAcuity        4       39221 457230 1190.4
## + System              2        1550 494900 1190.9
## + Temperature        3        6451 489999 1194.5
## - InputDevice         4     142428 638879 1194.7
## + Cautious            4       16031 480419 1196.8
##
## Step:  AIC=1176.87
## avg_time ~ Age + AvgSleepTime + HoursAweak + Distraction + Gamer +
##   CaffeinIntake + UsePrimaryHand + InputDevice + RefreshRate +
##   WiFi + Fatigue_3 + Stress_3 + Game_Frequency + Sports_Frequency +

```

```

##      Shooting_Levels
##
##      Df Sum of Sq    RSS    AIC
## - Sports_Frequency  1      917 497908 1172.2
## - Gamer              1     1223 498214 1172.3
## - Game_Frequency    1     2348 499339 1172.6
## - HoursAweak        1     2452 499443 1172.6
## - Shooting_Levels   2    26600 523591 1173.9
## - CaffeinIntake     1     9098 506089 1174.3
## - Distraction       1    10276 507267 1174.7
## - WiFi              1    12476 509467 1175.2
## - Fatigue_3         2    34112 531103 1175.7
## - UsePrimaryHand    1    15027 512017 1175.8
## <none>                496991 1176.9
## - RefreshRate       2    44203 541193 1178.1
## - Stress_3          2    47405 544396 1178.9
## + LastNightSleep    1      540 496451 1181.6
## + logAvgHoursExercise 1      75 496916 1181.7
## - AvgSleepTime      1    39943 536934 1182.0
## - Age               1    41602 538593 1182.4
## + Class             3    28221 468770 1183.9
## + Device            3    25335 471656 1184.7
## + PrimarilyHand     2     5551 491440 1185.1
## + Noise_Level       2     5306 491685 1185.2
## + VisualAcuity      4    38058 458933 1186.0
## + System            2     1434 495557 1186.2
## + Temperature       3     6038 490953 1189.9
## - InputDevice       4    146159 643150 1190.7
## + Cautious          4     15922 481069 1192.1
##
## Step:  AIC=1172.24
## avg_time ~ Age + AvgSleepTime + HoursAweak + Distraction + Gamer +
##      CaffeinIntake + UsePrimaryHand + InputDevice + RefreshRate +
##      WiFi + Fatigue_3 + Stress_3 + Game_Frequency + Shooting_Levels
##
##      Df Sum of Sq    RSS    AIC
## - Gamer              1     1386 499295 1167.7
## - HoursAweak        1     2721 500630 1168.1
## - Game_Frequency    1     3143 501051 1168.2
## - CaffeinIntake     1     8335 506243 1169.5
## - Shooting_Levels   2    28238 526146 1169.6
## - Distraction       1    10235 508143 1170.0
## - WiFi              1    12099 510007 1170.5
## - Fatigue_3         2    33464 531372 1170.9
## - UsePrimaryHand    1    15333 513241 1171.3
## <none>                497908 1172.2
## - RefreshRate       2    43934 541842 1173.4
## - Stress_3          2    49014 546922 1174.6
## + Sports_Frequency  1      917 496991 1176.9
## + LastNightSleep    1      500 497408 1177.0
## + logAvgHoursExercise 1      325 497583 1177.0
## - AvgSleepTime      1    39834 537743 1177.3
## - Age               1    42854 540762 1178.0
## + Class             3    27684 470225 1179.4

```

```

## + Device          3      25150 472758 1180.1
## + Noise_Level     2       5635 492273 1180.5
## + PrimarilyHand   2       4831 493077 1180.7
## + System          2       1516 496392 1181.6
## + VisualAcuity    4      37485 460423 1181.6
## + Temperature     3       5636 492273 1185.4
## - InputDevice     4     145248 643156 1185.8
## + Cautious        4       16828 481080 1187.2
##
## Step:  AIC=1167.74
## avg_time ~ Age + AvgSleepTime + HoursAweak + Distraction + CaffeinIntake +
##      UsePrimaryHand + InputDevice + RefreshRate + WiFi + Fatigue_3 +
##      Stress_3 + Game_Frequency + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - HoursAweak      1      2588 501882 1163.5
## - Game_Frequency   1      5180 504475 1164.2
## - CaffeinIntake     1     10350 509645 1165.5
## - Distraction       1     10866 510161 1165.7
## - Shooting_Levels   2     33060 532355 1166.3
## - Fatigue_3         2     33861 533156 1166.5
## - WiFi              1     14221 513515 1166.5
## - UsePrimaryHand    1     18935 518229 1167.7
## <none>              499295 1167.7
## - RefreshRate       2     43903 543198 1168.9
## - Stress_3          2     51385 550680 1170.7
## + Gamer             1      1386 497908 1172.2
## + Sports_Frequency   1      1081 498214 1172.3
## + LastNightSleep     1       503 498792 1172.5
## + logAvgHoursExercise 1       290 499005 1172.5
## - AvgSleepTime      1     39965 539260 1172.8
## - Age               1     43453 542747 1173.7
## + Class             3     28805 470489 1174.7
## + Device            3     25513 473781 1175.6
## + Noise_Level       2      6074 493221 1175.9
## + PrimarilyHand     2      5074 494220 1176.1
## + VisualAcuity      4     38863 460432 1176.7
## + System            2      1724 497571 1177.0
## + Temperature       3      6500 492795 1180.6
## + Cautious          4     15940 483355 1183.0
## - InputDevice       4     155703 654997 1183.3
##
## Step:  AIC=1163.55
## avg_time ~ Age + AvgSleepTime + Distraction + CaffeinIntake +
##      UsePrimaryHand + InputDevice + RefreshRate + WiFi + Fatigue_3 +
##      Stress_3 + Game_Frequency + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - Game_Frequency     1      4791 506674 1159.9
## - Distraction         1      9976 511858 1161.2
## - CaffeinIntake       1     10020 511902 1161.2
## - Fatigue_3           2     31583 533465 1161.7
## - WiFi                1     13625 515507 1162.2
## - Shooting_Levels     2     33587 535469 1162.2

```

```

## <none>                                501882 1163.5
## - UsePrimaryHand      1      20303 522185 1163.8
## - RefreshRate         2      41954 543837 1164.2
## - Stress_3            2      49054 550936 1165.9
## + HoursAweak          1       2588 499295 1167.7
## + Sports_Frequency    1       1354 500528 1168.1
## + Gamer               1       1253 500630 1168.1
## - AvgSleepTime        1      38386 540268 1168.2
## + LastNightSleep      1        528 501354 1168.3
## + logAvgHoursExercise 1        469 501413 1168.3
## - Age                 1      43493 545375 1169.4
## + Class               3      29478 472404 1170.3
## + PrimarilyHand       2       6881 495002 1171.5
## + Noise_Level         2       5836 496046 1171.8
## + Device              3      23758 478124 1171.9
## + VisualAcuity        4      38846 463036 1172.6
## + System              2       2226 499657 1172.7
## + Temperature         3       6485 495397 1176.5
## + Cautious            4      16785 485098 1178.6
## - InputDevice         4     156592 658474 1179.1
##
## Step: AIC=1159.92
## avg_time ~ Age + AvgSleepTime + Distraction + CaffeinIntake +
##      UsePrimaryHand + InputDevice + RefreshRate + WiFi + Fatigue_3 +
##      Stress_3 + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - Distraction      1      9399 516073 1157.4
## - Fatigue_3        2     30803 537476 1157.8
## - CaffeinIntake    1     11427 518101 1157.9
## - WiFi             1     17303 523976 1159.4
## <none>                                506674 1159.9
## - Shooting_Levels  2     40965 547638 1160.2
## - UsePrimaryHand   1     21635 528308 1160.5
## - RefreshRate      2     42254 548928 1160.5
## - Stress_3         2     52770 559444 1163.0
## + Game_Frequency   1      4791 501882 1163.5
## + Gamer            1      3135 503539 1164.0
## + Sports_Frequency 1      2730 503944 1164.1
## - AvgSleepTime     1     36788 543462 1164.1
## + HoursAweak       1      2199 504475 1164.2
## + logAvgHoursExercise 1      480 506193 1164.7
## + LastNightSleep   1       393 506281 1164.7
## - Age              1     47627 554301 1166.7
## + Class            3     28130 478544 1167.1
## + PrimarilyHand    2      7216 499457 1167.8
## + Noise_Level      2      5861 500812 1168.1
## + System           2      3106 503568 1168.8
## + VisualAcuity     4     39132 467542 1169.0
## + Device           3     20460 486214 1169.2
## + Temperature      3      5374 501300 1173.1
## + Cautious         4     15584 491090 1175.3
## - InputDevice      4     161112 667786 1176.1
##

```

```

## Step: AIC=1157.43
## avg_time ~ Age + AvgSleepTime + CaffeinIntake + UsePrimaryHand +
##      InputDevice + RefreshRate + WiFi + Fatigue_3 + Stress_3 +
##      Shooting_Levels
##
##      Df Sum of Sq    RSS    AIC
## - Fatigue_3      2    32434 548507 1155.6
## - WiFi           1    16084 532157 1156.5
## - CaffeinIntake   1    18303 534376 1157.1
## <none>                        516073 1157.4
## - Shooting_Levels 2    40953 557026 1157.6
## - RefreshRate     2    44636 560709 1158.4
## - UsePrimaryHand  1    25136 541209 1158.7
## + Distraction     1     9399 506674 1159.9
## - AvgSleepTime    1    34735 550808 1161.0
## + Game_Frequency  1     4215 511858 1161.2
## + Gamer           1     3809 512264 1161.3
## + Sports_Frequency 1     2540 513533 1161.7
## + HoursAweak      1     1426 514647 1161.9
## + logAvgHoursExercise 1     1099 514974 1162.0
## + LastNightSleep  1       167 515906 1162.2
## - Stress_3        2    62638 578711 1162.5
## - Age             1    45851 561924 1163.5
## + Class           3    29900 486173 1164.3
## + VisualAcuity    4    43461 472612 1165.5
## + PrimarilyHand   2     5354 510719 1165.8
## + Device          3    23544 492529 1166.0
## + System          2     4423 511650 1166.0
## + Noise_Level     2     3899 512174 1166.2
## + Temperature     3     5094 510979 1170.7
## + Cautious        4    17788 498285 1172.3
## - InputDevice     4   170658 686731 1174.8
##
## Step: AIC=1155.57
## avg_time ~ Age + AvgSleepTime + CaffeinIntake + UsePrimaryHand +
##      InputDevice + RefreshRate + WiFi + Stress_3 + Shooting_Levels
##
##      Df Sum of Sq    RSS    AIC
## - CaffeinIntake   1    14370 562877 1154.0
## - WiFi           1    18603 567110 1155.0
## <none>                        548507 1155.6
## - RefreshRate     2    49010 597517 1156.9
## - Stress_3        2    50136 598643 1157.1
## + Fatigue_3       2    32434 516073 1157.4
## + Distraction     1    11031 537476 1157.8
## - UsePrimaryHand  1    33482 581989 1158.3
## - Shooting_Levels 2    57337 605844 1158.7
## + Gamer           1     4553 543954 1159.4
## + Game_Frequency  1     3318 545189 1159.7
## + LastNightSleep  1     2410 546097 1159.9
## - AvgSleepTime    1    41446 589953 1160.1
## + HoursAweak      1      952 547555 1160.2
## + Sports_Frequency 1       875 547632 1160.2
## - Age             1    42449 590956 1160.3

```

```

## + logAvgHoursExercise 1      194 548313 1160.4
## + PrimarilyHand      2      7688 540819 1163.5
## + VisualAcuity       4     46810 501697 1163.5
## + Class              3     25855 522652 1163.9
## + System             2      4806 543701 1164.2
## + Device             3     23110 525397 1164.6
## + Noise_Level        2      1086 547421 1165.0
## + Temperature       3      8404 540103 1168.2
## + Cautious           4     23783 524724 1169.3
## - InputDevice        4    164875 713382 1170.0
##
## Step:  AIC=1154.05
## avg_time ~ Age + AvgSleepTime + UsePrimaryHand + InputDevice +
## RefreshRate + WiFi + Stress_3 + Shooting_Levels
##
##              Df Sum of Sq  RSS    AIC
## - WiFi              1    16704 579581 1153.0
## <none>                  562877 1154.0
## - RefreshRate       2    47383 610260 1154.8
## + Distraction        1    17472 545405 1154.8
## - UsePrimaryHand     1    28044 590921 1155.5
## + CaffeinIntake      1    14370 548507 1155.6
## - Stress_3          2    54950 617826 1156.3
## - Shooting_Levels    2    56682 619558 1156.7
## + Gamer             1     8444 554433 1157.0
## + Fatigue_3          2    28501 534376 1157.1
## + Game_Frequency     1     4306 558570 1157.9
## - AvgSleepTime       1    42193 605070 1158.5
## + LastNightSleep     1     1688 561188 1158.5
## + logAvgHoursExercise 1     1178 561698 1158.6
## + HoursAweak         1      909 561968 1158.7
## + Sports_Frequency   1      96 562781 1158.9
## - Age               1    57490 620367 1161.7
## + System            2     6116 556760 1162.4
## + Device            3    26692 536185 1162.4
## + PrimarilyHand      2     6096 556781 1162.4
## + Noise_Level        2     1298 561579 1163.5
## + Class              3    19876 543000 1164.0
## + VisualAcuity       4    39616 523260 1164.1
## + Temperature       3    10323 552554 1166.2
## + Cautious           4    17666 545210 1169.4
## - InputDevice        4   186676 749553 1171.6
##
## Step:  AIC=1152.96
## avg_time ~ Age + AvgSleepTime + UsePrimaryHand + InputDevice +
## RefreshRate + Stress_3 + Shooting_Levels
##
##              Df Sum of Sq  RSS    AIC
## - RefreshRate       2    38588 618169 1151.5
## <none>                  579581 1153.0
## - Stress_3          2    46025 625606 1153.1
## - UsePrimaryHand     1    25145 604725 1153.6
## - Shooting_Levels    2    50150 629731 1154.0
## + WiFi              1    16704 562877 1154.0

```

```

## + Distraction      1      15752 563829 1154.3
## + Gamer            1      12747 566834 1155.0
## + CaffeinIntake    1      12471 567110 1155.0
## + Fatigue_3        2      31821 547760 1155.4
## + Game_Frequency   1       7149 572432 1156.2
## - AvgSleepTime     1      39927 619508 1156.7
## + logAvgHoursExercise 1       1643 577938 1157.5
## + HoursAweak       1       1097 578484 1157.6
## + LastNightSleep   1       552 579029 1157.7
## + Sports_Frequency 1        65 579516 1157.8
## - Age              1      56084 635665 1160.0
## + System            2       8240 571341 1160.8
## + Device            3      28221 551360 1161.1
## + PrimarilyHand     2       5849 573732 1161.4
## + Noise_Level       2       1237 578344 1162.4
## + Class              3      20137 559444 1163.0
## + VisualAcuity      4      37486 542095 1163.8
## + Temperature       3      12005 567576 1164.8
## + Cautious          4      17982 561599 1168.3
## - InputDevice       4     206012 785593 1172.8
##
## Step:  AIC=1151.55
## avg_time ~ Age + AvgSleepTime + UsePrimaryHand + InputDevice +
##      Stress_3 + Shooting_Levels
##
##              Df Sum of Sq    RSS    AIC
## - Stress_3      2     31408 649578 1148.2
## - UsePrimaryHand 1     20059 638228 1150.8
## <none>                      618169 1151.5
## - AvgSleepTime   1     28131 646300 1152.4
## + Distraction     1     18466 599703 1152.5
## + RefreshRate     2     38588 579581 1153.0
## + Gamer           1     12207 605962 1153.8
## + CaffeinIntake   1     11635 606535 1154.0
## + Fatigue_3       2     32187 585983 1154.4
## + WiFi            1       7909 610260 1154.8
## + Game_Frequency  1       6661 611508 1155.0
## + logAvgHoursExercise 1      5245 612925 1155.3
## + HoursAweak      1       3800 614370 1155.6
## - Shooting_Levels 2     70043 688212 1155.7
## + LastNightSleep  1        803 617367 1156.2
## + Sports_Frequency 1         16 618153 1156.4
## + Device          3     41828 576341 1157.1
## - Age             1     58968 677138 1158.5
## + System          2       9841 608328 1159.2
## + PrimarilyHand   2       2122 616047 1160.8
## + Noise_Level     2        333 617836 1161.2
## + Class           3     17642 600527 1162.4
## + VisualAcuity    4     38691 579479 1162.7
## + Temperature     3     16185 601985 1162.7
## + Cautious        4     12811 605358 1168.3
## - InputDevice     4    271562 889731 1179.1
##
## Step:  AIC=1148.23

```

```

## avg_time ~ Age + AvgSleepTime + UsePrimaryHand + InputDevice +
##   Shooting_Levels
##
##           Df Sum of Sq    RSS    AIC
## - UsePrimaryHand      1    11011 660588 1145.5
## + Distraction          1    28950 620627 1147.2
## - AvgSleepTime         1    24124 673701 1148.1
## <none>                  649578 1148.2
## + Gamer                1    16505 633072 1149.8
## + CaffeinIntake         1    15521 634056 1150.0
## + Game_Frequency        1     8352 641225 1151.4
## + Stress_3              2    31408 618169 1151.5
## + HoursAweak            1     4977 644601 1152.1
## + WiFi                  1     3966 645611 1152.3
## + logAvgHoursExercise   1     1322 648256 1152.8
## + Sports_Frequency      1      647 648930 1153.0
## + LastNightSleep        1       20 649558 1153.1
## + RefreshRate           2    23972 625606 1153.1
## + Device                3    45013 604564 1153.5
## + Fatigue_3             2    21485 628092 1153.6
## - Shooting_Levels       2    81570 731148 1153.8
## + System                2    11376 638202 1155.7
## + Class                 3    33514 616064 1156.0
## + PrimarilyHand         2     3633 645945 1157.2
## - Age                   1    75656 725234 1157.6
## + Noise_Level           2     1087 648490 1157.7
## + Temperature           3    12004 637574 1160.4
## + VisualAcuity          4    28589 620989 1161.9
## + Cautious              4    18355 631223 1164.0
## - InputDevice           4    261468 911045 1172.4
##
## Step:  AIC=1145.54
## avg_time ~ Age + AvgSleepTime + InputDevice + Shooting_Levels
##
##           Df Sum of Sq    RSS    AIC
## + Distraction          1    29814 630774 1144.4
## - AvgSleepTime         1    21999 682587 1144.9
## <none>                  660588 1145.5
## + Gamer                1    21720 638869 1146.1
## + CaffeinIntake         1    11642 648946 1148.1
## + UsePrimaryHand        1    11011 649578 1148.2
## + Game_Frequency        1     8848 651740 1148.7
## - Shooting_Levels       2    73292 733880 1149.4
## + HoursAweak            1     4211 656378 1149.6
## + WiFi                  1     3701 656887 1149.7
## + logAvgHoursExercise   1     1219 659369 1150.2
## + Sports_Frequency      1      925 659663 1150.2
## + Fatigue_3             2    25158 635430 1150.2
## + LastNightSleep        1       2 660586 1150.4
## + RefreshRate           2    22553 638035 1150.8
## + Stress_3              2    22360 638228 1150.8
## + Device                3    38732 621856 1152.3
## + Class                 3    35674 624914 1153.0
## + System                2    10872 649716 1153.1

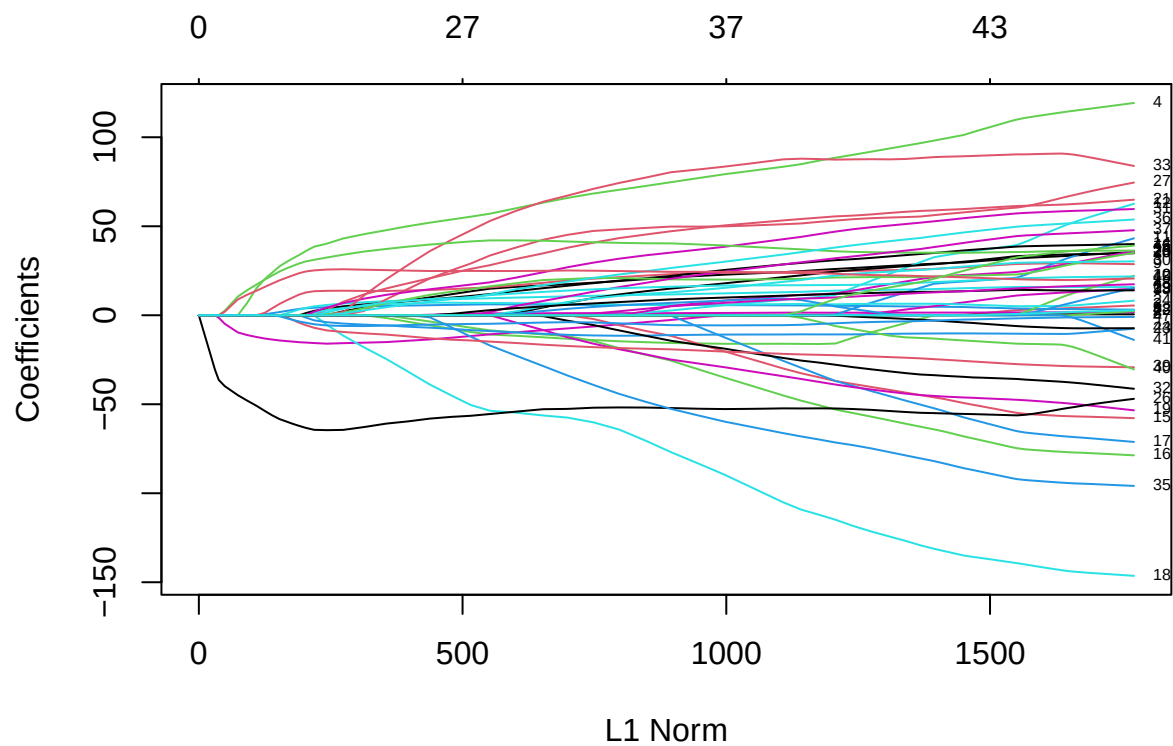
```

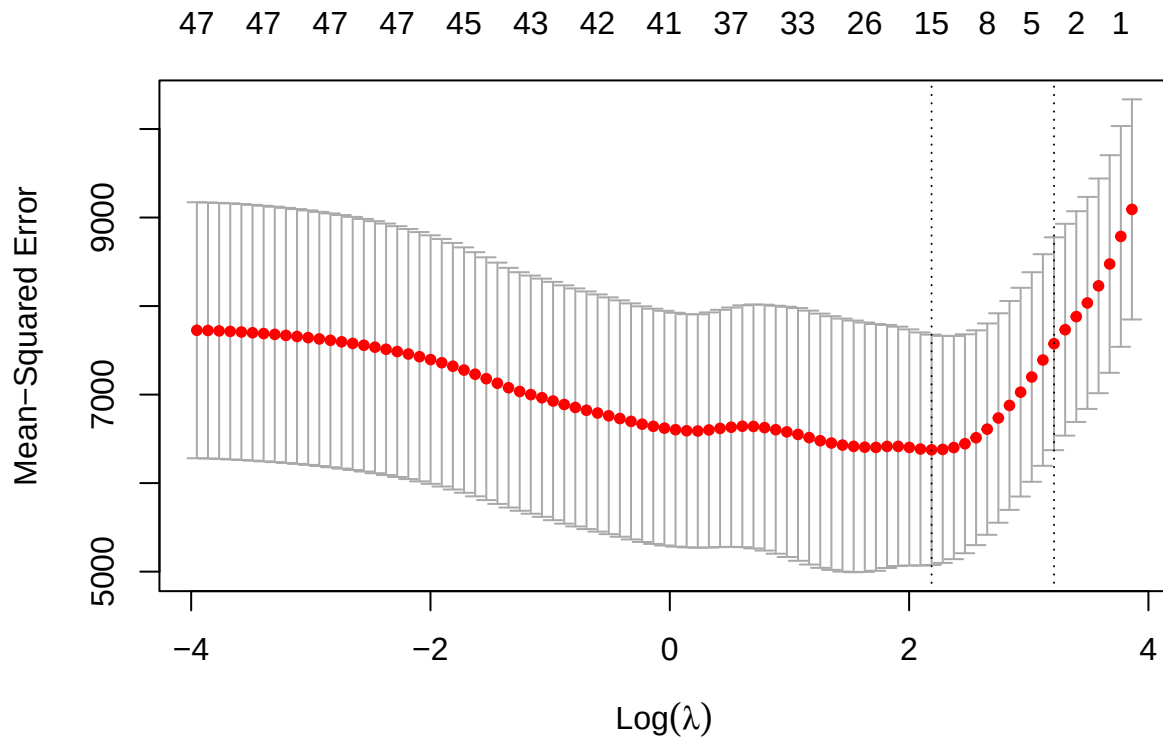
```

## + PrimarilyHand      2      2970 657618 1154.7
## + Noise_Level        2        307 660281 1155.2
## - Age                1      80964 741552 1155.6
## + Temperature        3      13189 647399 1157.5
## + VisualAcuity       4      31287 629301 1158.7
## + Cautious           4      19194 641394 1161.2
## - InputDevice        4     276674 937262 1171.2
##
## Step: AIC=1144.44
## avg_time ~ Age + AvgSleepTime + InputDevice + Shooting_Levels +
##   Distraction
##
##              Df Sum of Sq    RSS    AIC
## <none>                        630774 1144.4
## - AvgSleepTime      1      26881 657655 1145.0
## - Distraction       1      29814 660588 1145.5
## + Gamer             1      16241 614533 1145.9
## + UsePrimaryHand    1      10147 620627 1147.2
## + Game_Frequency    1      10002 620772 1147.2
## - Shooting_Levels   2      66798 697572 1147.7
## + WiFi              1       5756 625018 1148.1
## + CaffeinIntake     1       4188 626586 1148.4
## + logAvgHoursExercise 1       2644 628130 1148.8
## + HoursAweak        1       2039 628735 1148.9
## + Fatigue_3         2      24828 605946 1149.0
## + Sports_Frequency  1       1474 629300 1149.0
## + LastNightSleep    1        333 630441 1149.2
## + RefreshRate       2      21687 609087 1149.7
## + Stress_3          2      14083 616691 1151.2
## + Device            3      31374 599400 1152.4
## + Class             3      30755 600019 1152.6
## + Noise_Level       2       6423 624351 1152.8
## + System            2       6359 624415 1152.8
## + PrimarilyHand     2       3173 627601 1153.5
## - Age              1      75201 705976 1154.1
## + Temperature       3      14039 616735 1156.1
## + VisualAcuity      4      27428 603346 1158.1
## + Cautious          4      14802 615972 1160.8
## - InputDevice       4     247894 878668 1167.8
##
##
## Call:
## lm(formula = avg_time ~ Age + AvgSleepTime + InputDevice + Shooting_Levels +
##   Distraction, data = Survey_cleaned)
##
## Coefficients:
##              (Intercept)                      Age
##                113.638                      15.768
##              AvgSleepTime              InputDeviceMouse
##                14.512                  -88.056
##              InputDeviceKeyboard              InputDeviceTrackpad
##                56.524                      8.402
##              InputDeviceTouch screen  Shooting_LevelsIntermediate Levels
##                -28.410                  22.652

```

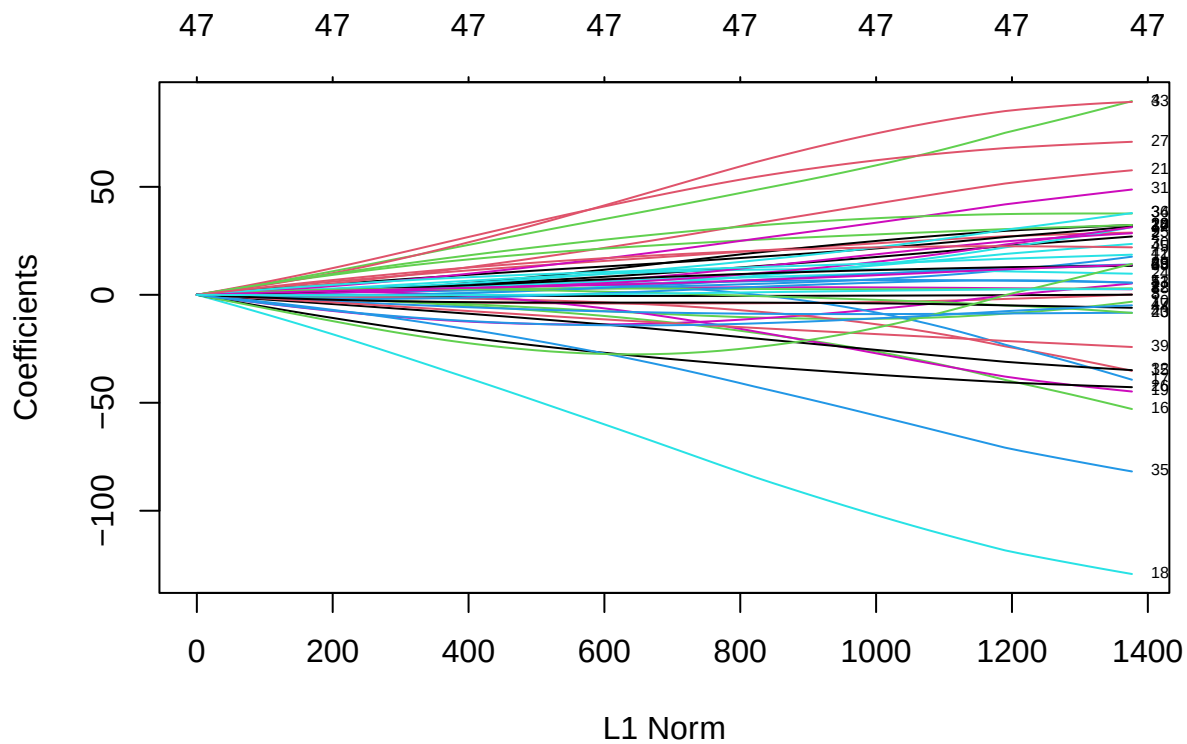
##	Shooting_LevelsLow Levels	DistractionYes
##	58.928	37.854

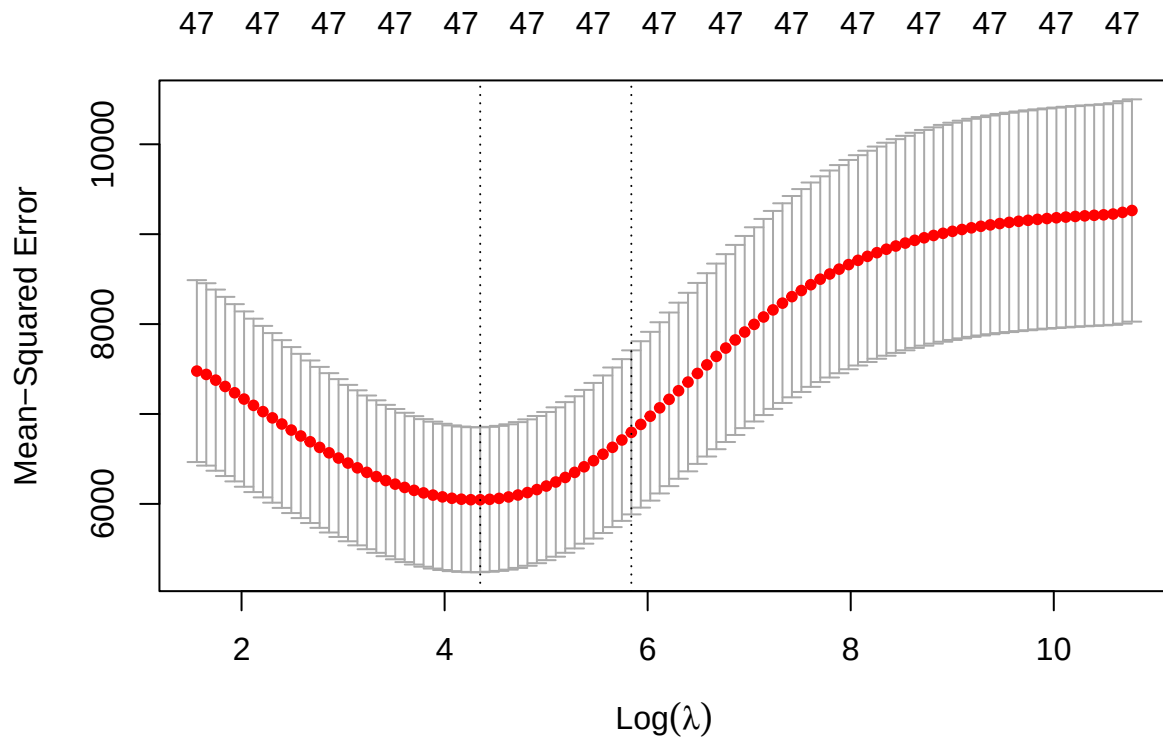




```
## 49 x 1 sparse Matrix of class "dgCMatrix"
##
## (Intercept) 434.637178
## (Intercept) .
## ClassJunior .
## ClassSenior .
## ClassGraduate student 38.495659
## Age 4.207794
## AvgSleepTime 4.959268
## LastNightSleep .
## HoursAweek .
## DistractionYes 13.502323
## TemperatureWarm .
## TemperatureNeutral .
## TemperatureCold .
## GamerYes -15.617365
## CaffeinIntakeYes 2.454916
## VisualAcuityExcellent .
## VisualAcuityGood .
## VisualAcuityPoor .
## VisualAcuityVery Poor .
## PrimarilyHandLeft hand .
## PrimarilyHandAmbidextrous (both hands equally) .
## UsePrimaryHandNo .
## CautiousVery cautious .
## CautiousModerately cautious -2.849909
```

## CautiousSlightly cautious	1.386369
## CautiousNot cautious at all	.
## InputDeviceMouse	-64.358406
## InputDeviceKeyboard	.
## InputDeviceTrackpad	0.167877
## InputDeviceTouch screen	.
## RefreshRateMid-Range	.
## RefreshRateStandard	2.010429
## DeviceiPad(Tablet	.
## DeviceiPhone(Smartphone	.
## DeviceLaptop	31.126876
## WiFiStable	.
## Fatigue_3Moderately Fatigue	.
## Fatigue_3Low Fatigue	.
## Stress_3Moderately Stress	.
## Stress_3Low Stress	-7.020384
## Noise_LevelModerately Noise	.
## Noise_LevelLow Noise	.
## Game_FrequencyLow Frequency	.
## Sports_FrequencyLow Frequency	.
## Shooting_LevelsIntermediate Levels	.
## Shooting_LevelsLow Levels	25.202538
## SystemLinux	.
## SystemWindows	-5.059475
## logAvgHoursExercise	.





```
## 49 x 1 sparse Matrix of class "dgCMatrix"
##
## (Intercept) 368.5377698
## (Intercept) .
## ClassJunior 9.2179213
## ClassSenior -4.0525003
## ClassGraduate student 37.7032688
## Age 5.1690070
## AvgSleepTime 6.5175895
## LastNightSleep 3.1701663
## HoursAweek -0.5404607
## DistractionYes 16.6195321
## TemperatureWarm -8.2325000
## TemperatureNeutral 3.5619055
## TemperatureCold 4.6298805
## GamerYes -13.5314439
## CaffeinIntakeYes 13.9420796
## VisualAcuityExcellent -4.0959600
## VisualAcuityGood -11.0979381
## VisualAcuityPoor 4.1943588
## VisualAcuityVery Poor -64.8507779
## PrimarilyHandLeft hand -8.0711916
## PrimarilyHandAmbidextrous (both hands equally) 13.1496551
## UsePrimaryHandNo 23.8720903
## CautiousVery cautious 2.9004930
## CautiousModerately cautious -7.9892714
```

```

## CautiousSlightly cautious          10.2090207
## CautiousNot cautious at all        7.5536393
## InputDeviceMouse                   -28.1947413
## InputDeviceKeyboard                43.8645305
## InputDeviceTrackpad                22.3201868
## InputDeviceTouch screen            3.9222785
## RefreshRateMid-Range               6.6727554
## RefreshRateStandard                18.5185918
## DeviceiPad(Tablet                  -14.9492712
## DeviceiPhone(Smartphone            45.4042015
## DeviceLaptop                       26.9626098
## WiFiStable                         -29.9684480
## Fatigue_3Moderately Fatigue        10.8075295
## Fatigue_3Low Fatigue               6.1120911
## Stress_3Moderately Stress           7.7363361
## Stress_3Low Stress                 -12.1926836
## Noise_LevelModerately Noise         1.0909481
## Noise_LevelLow Noise               1.2814921
## Game_FrequencyLow Frequency        11.0270908
## Sports_FrequencyLow Frequency       4.9663124
## Shooting_LevelsIntermediate Levels -3.6278837
## Shooting_LevelsLow Levels          17.7413953
## SystemLinux                        -27.5803947
## SystemWindows                      -14.1677818
## logAvgHoursExercise                0.4618736

##
## Call:
## lm(formula = avg_time ~ Class + AvgSleepTime + CaffeinIntake +
##     VisualAcuity + PrimarilyHand + UsePrimaryHand + InputDevice +
##     RefreshRate + Device + WiFi + Fatigue_3 + Stress_3 + Game_Frequency,
##     data = Survey_cleaned)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -138.894  -31.697   -6.822   33.277  192.712
##
## Coefficients:
##                                     Estimate Std. Error t value
## (Intercept)                       425.317    105.952    4.014
## ClassJunior                        38.217     15.069    2.536
## ClassSenior                        13.981     18.142    0.771
## ClassGraduate student             130.035     28.253    4.602
## AvgSleepTime                       15.003      6.128    2.448
## CaffeinIntakeYes                   39.404     14.849    2.654
## VisualAcuityExcellent              -69.883     25.778   -2.711
## VisualAcuityGood                  -82.245     26.164   -3.143
## VisualAcuityPoor                  -80.451     42.878   -1.876
## VisualAcuityVery Poor             -117.412     75.133   -1.563
## PrimarilyHandLeft hand             -41.362     35.280   -1.172
## PrimarilyHandAmbidextrous (both hands equally) 32.110     36.761    0.873
## UsePrimaryHandNo                   71.433     33.181    2.153
## InputDeviceMouse                  -31.564     68.331   -0.462
## InputDeviceKeyboard                68.597     85.193    0.805

```

```

## InputDeviceTrackpad          45.508      67.390      0.675
## InputDeviceTouch screen      -3.979      75.914     -0.052
## RefreshRateMid-Range         28.670      19.571      1.465
## RefreshRateStandard          55.821      18.704      2.984
## DeviceiPad(Tablet            -11.942      53.720     -0.222
## DeviceiPhone(Smartphone      140.629      63.237      2.224
## DeviceLaptop                 38.479      21.151      1.819
## WiFiStable                   -65.223      49.872     -1.308
## Fatigue_3Moderately Fatigue   62.065      20.502      3.027
## Fatigue_3Low Fatigue          54.603      20.072      2.720
## Stress_3Moderately Stress      7.351      17.207      0.427
## Stress_3Low Stress           -35.489      18.291     -1.940
## Game_FrequencyLow Frequency   26.502      12.463      2.127
##                               Pr(>|t|)
## (Intercept)                  0.000115 ***
## ClassJunior                   0.012744 *
## ClassSenior                   0.442730
## ClassGraduate student        1.22e-05 ***
## AvgSleepTime                 0.016081 *
## CaffeinIntakeYes             0.009250 **
## VisualAcuityExcellent        0.007885 **
## VisualAcuityGood             0.002193 **
## VisualAcuityPoor             0.063504 .
## VisualAcuityVery Poor        0.121248
## PrimarilyHandLeft hand       0.243804
## PrimarilyHandAmbidextrous (both hands equally) 0.384468
## UsePrimaryHandNo             0.033711 *
## InputDeviceMouse             0.645123
## InputDeviceKeyboard          0.422595
## InputDeviceTrackpad          0.501031
## InputDeviceTouch screen      0.958303
## RefreshRateMid-Range         0.146044
## RefreshRateStandard          0.003564 **
## DeviceiPad(Tablet            0.824536
## DeviceiPhone(Smartphone      0.028387 *
## DeviceLaptop                 0.071840 .
## WiFiStable                   0.193907
## Fatigue_3Moderately Fatigue   0.003131 **
## Fatigue_3Low Fatigue          0.007680 **
## Stress_3Moderately Stress     0.670124
## Stress_3Low Stress            0.055140 .
## Game_FrequencyLow Frequency   0.035894 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 64.09 on 101 degrees of freedom
## Multiple R-squared:  0.6454, Adjusted R-squared:  0.5506
## F-statistic: 6.808 on 27 and 101 DF,  p-value: 4.733e-13

##                               GVIF Df GVIF^(1/(2*Df))
## Class                1.882939  3      1.111235
## AvgSleepTime         1.242119  1      1.114504
## CaffeinIntake         1.291534  1      1.136457
## VisualAcuity          1.918683  4      1.084864

```

```
## PrimarilyHand 2.157614 2 1.211974
## UsePrimaryHand 1.774313 1 1.332033
## InputDevice 6.483538 4 1.263213
## RefreshRate 2.266885 2 1.227036
## Device 6.241066 3 1.356885
## WiFi 1.192216 1 1.091886
## Fatigue_3 1.749256 2 1.150041
## Stress_3 1.897225 2 1.173626
## Game_Frequency 1.198209 1 1.094627
```

```
## Analysis of Variance Table
##
```

```
## Response: avg_time
```

```
##          Df Sum Sq Mean Sq F value    Pr(>F)
## Class          3  70764    23588   4.6315 0.0047165 **
## AvgSleepTime    1  51358    51358  10.0842 0.0020692 **
## CaffeinIntake    1  44474    44474   8.7324 0.0040213 **
## VisualAcuity     4 112552    28138   5.5249 0.0005194 ***
## UsePrimaryHand    1   8500     8500   1.6689 0.1998229
## RefreshRate      2 133463    66731  13.1028 1.061e-05 ***
## Device           3  90243    30081   5.9064 0.0010229 **
## WiFi            1  19165    19165   3.7631 0.0556363 .
## Fatigue_3        2  29443    14722   2.8906 0.0608947 .
## Stress_3         2  33627    16814   3.3014 0.0414977 *
## Game_Frequency    1  31181    31181   6.1224 0.0152893 *
## Class:AvgSleepTime 3   4805     1602   0.3145 0.8148506
## AvgSleepTime:CaffeinIntake 1   2820     2820   0.5538 0.4587899
## AvgSleepTime:VisualAcuity 3   4600     1533   0.3011 0.8245341
## AvgSleepTime:PrimarilyHand 2  19859     9929   1.9497 0.1484902
## AvgSleepTime:UsePrimaryHand 1   4988     4988   0.9794 0.3250906
## AvgSleepTime:RefreshRate 2  21825    10912   2.1426 0.1234963
## AvgSleepTime:Device 2  20285    10143   1.9915 0.1426606
## AvgSleepTime:WiFi 1    162      162   0.0318 0.8588795
## AvgSleepTime:Fatigue_3 2  13297     6648   1.3054 0.2763119
## AvgSleepTime:Stress_3 2   8767     4383   0.8607 0.4264506
## AvgSleepTime:Game_Frequency 1    694      694   0.1363 0.7129253
## Residuals       87 443084     5093
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## Analysis of Variance Table
##
```

```
## Response: avg_time
```

```
##          Df Sum Sq Mean Sq F value    Pr(>F)
## Class          3  70764    23588   4.6294 0.0043877 **
## AvgSleepTime    1  51358    51358  10.0797 0.0019583 **
## CaffeinIntake    1  44474    44474   8.7285 0.0038528 **
## VisualAcuity     4 112552    28138   5.5225 0.0004426 ***
## UsePrimaryHand    1   8500     8500   1.6682 0.1992846
## RefreshRate      2 133463    66731  13.0969 8.168e-06 ***
## Device           3  90243    30081   5.9038 0.0009104 ***
## WiFi            1  19165    19165   3.7614 0.0550824 .
## Fatigue_3        2  29443    14722   2.8893 0.0599655 .
## Stress_3         2  33627    16814   3.2999 0.0406746 *
```

```

## Game_Frequency    1  31181    31181  6.1196 0.0149411 *
## Residuals         107 545186    5095
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## Analysis of Variance Table
##
## Response: avg_time
##              Df Sum Sq Mean Sq F value    Pr(>F)
## Class          3  70764   23588   4.4860 0.0052020 **
## AvgSleepTime    1  51358   51358   9.7673 0.0022721 **
## CaffeinIntake    1  44474   44474   8.4581 0.0043975 **
## VisualAcuity     4 112552   28138   5.3513 0.0005638 ***
## RefreshRate      2 124040   62020 11.7950 2.286e-05 ***
## Device           3  82181   27394   5.2098 0.0021120 **
## Fatigue_3        2  39237   19619   3.7311 0.0270521 *
## Game_Frequency    1  44518   44518   8.4664 0.0043787 **
## UsePrimaryHand    1  22436   22436   4.2668 0.0412128 *
## Residuals       110 578395    5258
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##              GVIF Df GVIF^(1/(2*Df))
## Age              1.083396  1      1.040863
## AvgSleepTime     1.064404  1      1.031699
## InputDevice       1.183969  4      1.021333
## Shooting_Levels  1.095841  2      1.023144
## Distraction       1.053618  1      1.026459

##
## Call:
## lm(formula = avg_time ~ Age + AvgSleepTime + InputDevice + Shooting_Levels +
##     Distraction, data = Survey_cleaned)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -128.316  -49.446   -2.732   51.405  201.703
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    113.638    124.733   0.911  0.364110
## Age              15.768     4.186   3.767  0.000259 ***
## AvgSleepTime    14.512     6.444   2.252  0.026159 *
## InputDeviceMouse -88.056    74.274  -1.186  0.238156
## InputDeviceKeyboard  56.524    89.861   0.629  0.530541
## InputDeviceTrackpad  8.402    73.690   0.114  0.909417
## InputDeviceTouch screen -28.410    76.999  -0.369  0.712813
## Shooting_LevelsIntermediate Levels  22.652    20.549   1.102  0.272535
## Shooting_LevelsLow Levels    58.928    18.504   3.185  0.001851 **
## DistractionYes    37.854    15.961   2.372  0.019315 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##

```

```
## Residual standard error: 72.81 on 119 degrees of freedom
## Multiple R-squared:  0.4609, Adjusted R-squared:  0.4201
## F-statistic: 11.3 on 9 and 119 DF,  p-value: 1.231e-12
```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: avg_time
```

```
##
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
## Age	1	37863	37863	5.9245	0.0164430 *
## AvgSleepTime	1	60340	60340	9.4417	0.0026391 **
## Shooting_Levels	2	134489	67245	10.5220	6.278e-05 ***
## Distraction	1	58594	58594	9.1684	0.0030297 **
## Age:Shooting_Levels	2	120026	60013	9.3905	0.0001651 ***
## Age:Distraction	1	1556	1556	0.2435	0.6225874
## AvgSleepTime:Shooting_Levels	2	9066	4533	0.7093	0.4941076
## AvgSleepTime:Distraction	1	288	288	0.0451	0.8321666
## Residuals	117	747731	6391		

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: avg_time
```

```
##
```

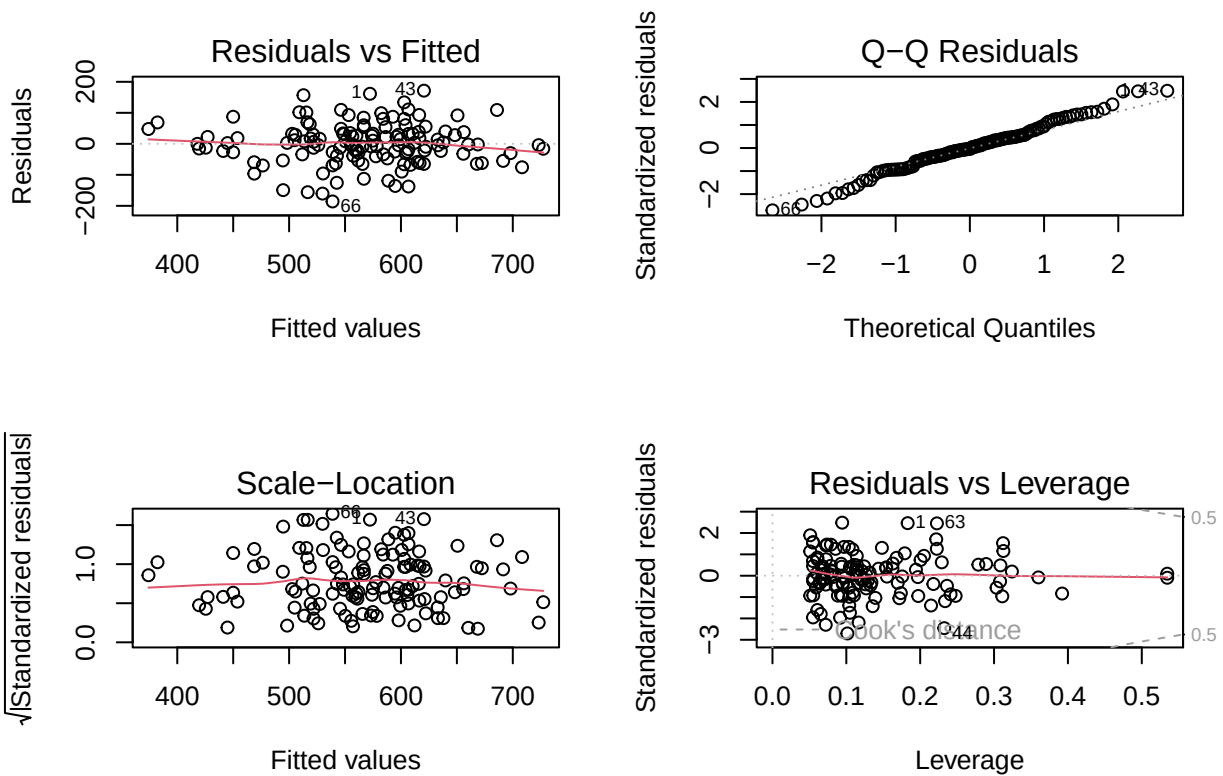
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
## Age	1	37863	37863	6.0390	0.0154107 *
## AvgSleepTime	1	60340	60340	9.6240	0.0023906 **
## Shooting_Levels	2	134489	67245	10.7252	5.151e-05 ***
## Distraction	1	58594	58594	9.3455	0.0027516 **
## Age:Shooting_Levels	2	120026	60013	9.5718	0.0001383 ***
## Residuals	121	758642	6270		

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## Warning: not plotting observations with leverage one:
```

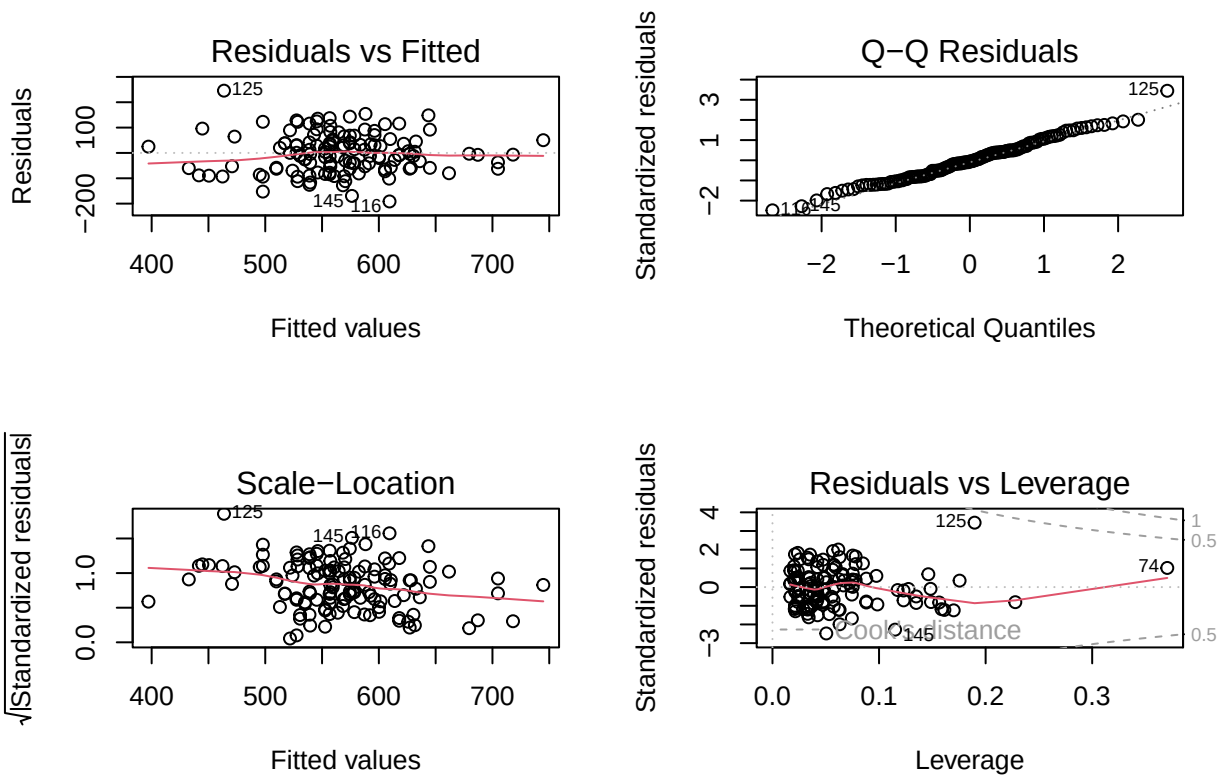
```
## 14
```



```
##
## studentized Breusch-Pagan test
##
## data: AICmod4
## BP = 26.426, df = 18, p-value = 0.0904

##
## Shapiro-Wilk normality test
##
## data: residuals(AICmod4)
## W = 0.98837, p-value = 0.3476

##
## Durbin-Watson test
##
## data: AICmod4
## DW = 2.1674, p-value = 0.8238
## alternative hypothesis: true autocorrelation is greater than 0
```



```
##
## studentized Breusch-Pagan test
##
## data: BICmod3
## BP = 18.42, df = 7, p-value = 0.01021

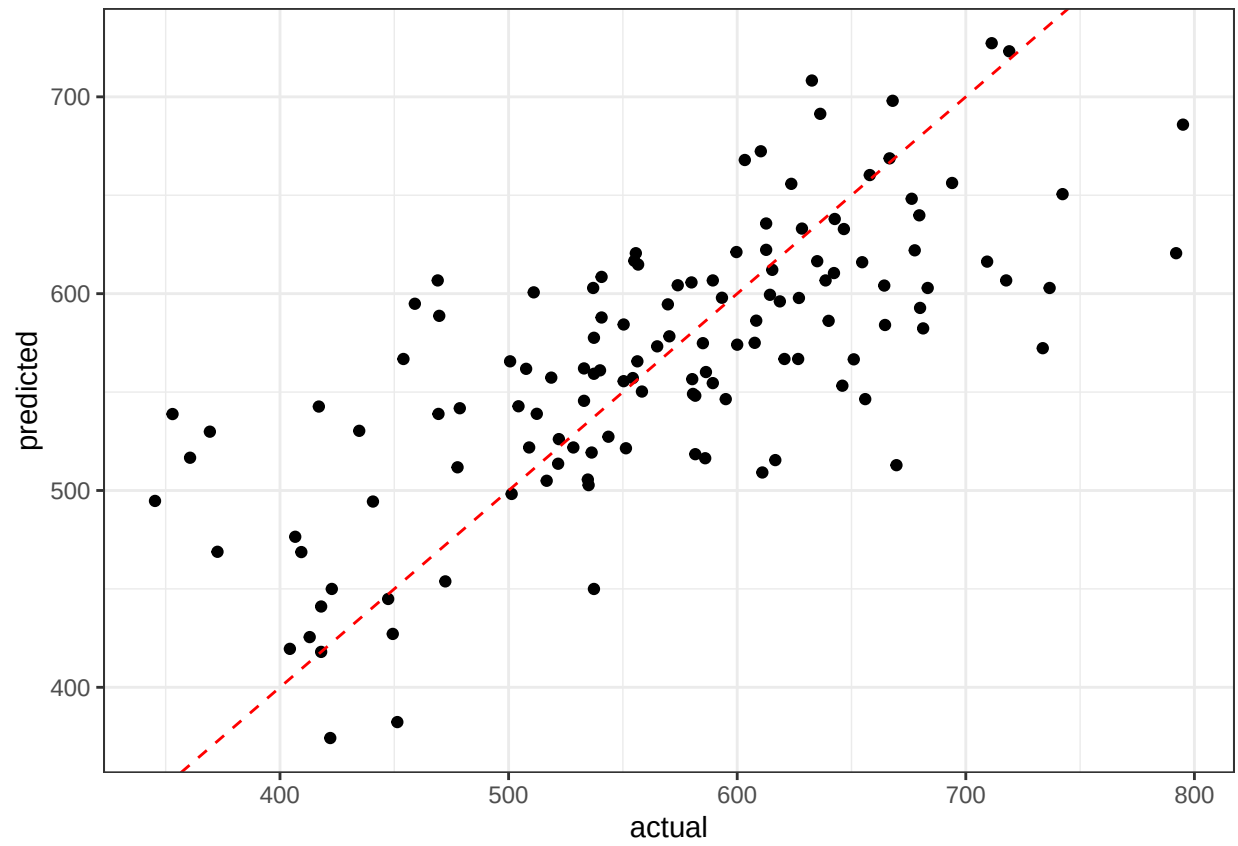
##
## Shapiro-Wilk normality test
##
## data: residuals(BICmod3)
## W = 0.99066, p-value = 0.5411

##
## Durbin-Watson test
##
## data: BICmod3
## DW = 1.649, p-value = 0.02156
## alternative hypothesis: true autocorrelation is greater than 0

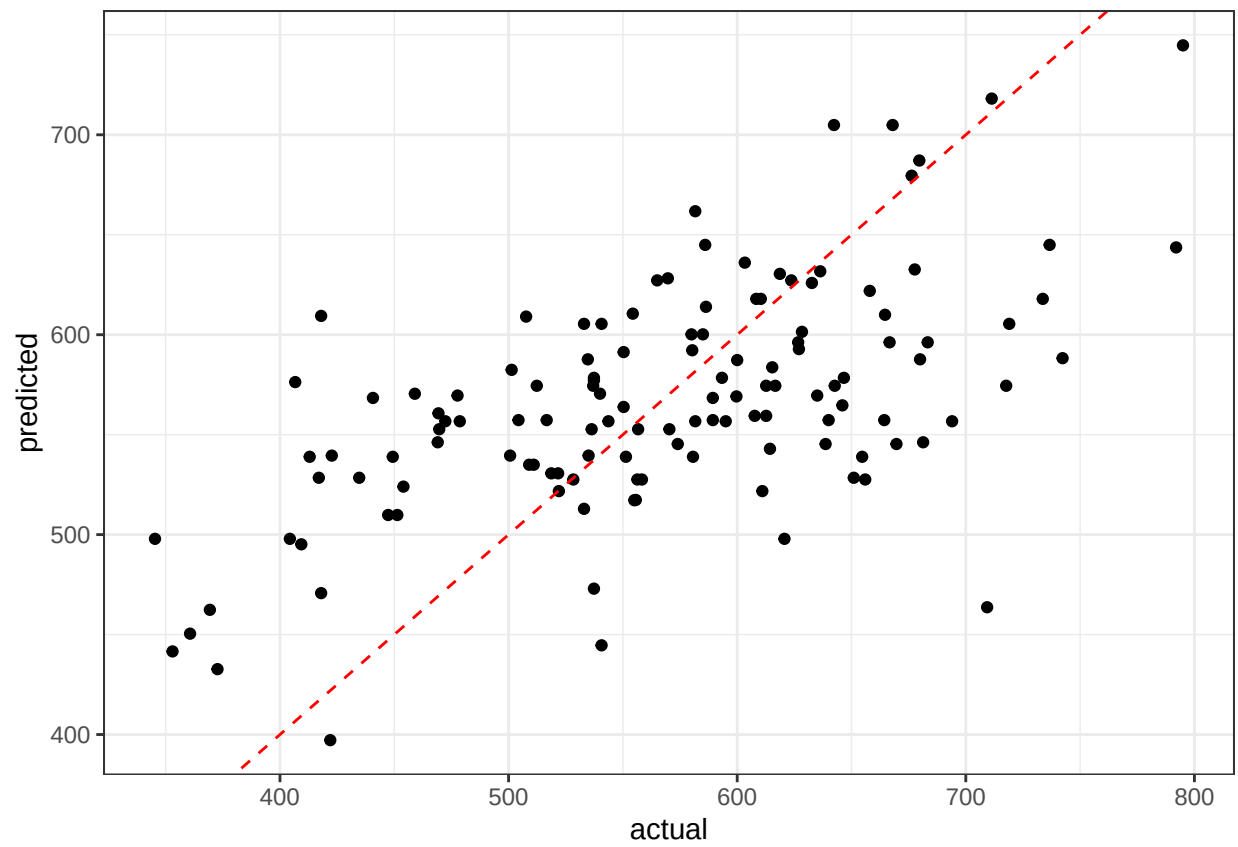
##
## studentized Breusch-Pagan test
##
## data: BICmod3_weighted
## BP = 9.2754, df = 7, p-value = 0.2335
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: residuals(BICmod3_weighted)  
## W = 0.99034, p-value = 0.511
```

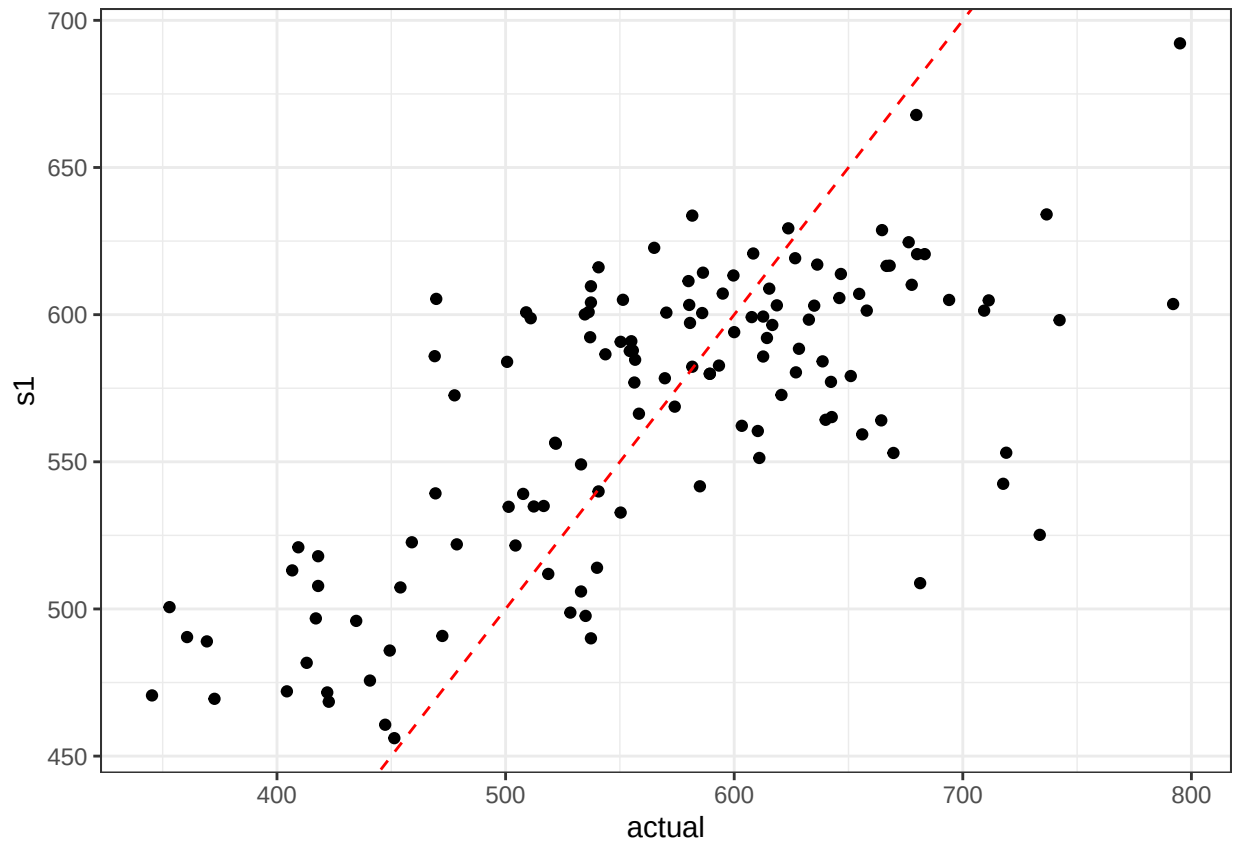
```
## [1] 66.96031
```



```
## [1] 76.78642
```



```
## [1] 70.01466
```



```
## [1] 62.5103
```

